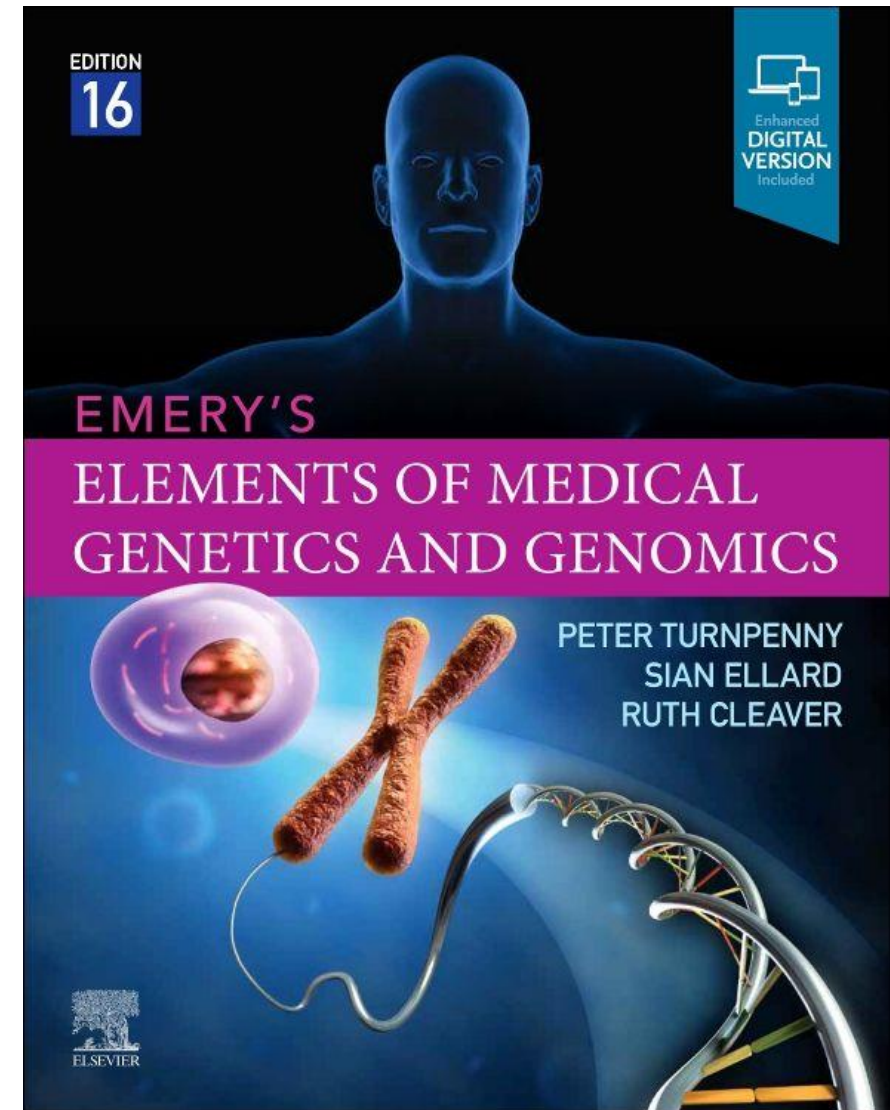


Mendelian Inheritance

Dr. Dalal Al-Shaer
Assistant professor



- **Chapter 6**
- **Page 66-75**



Objectives:

- 1- List the different Mendelin inheritance patterns.
- 2- Draw a pedigree for a given family and label it.
- 3- List the different factors that affect the phenotype.
- 4- List some clinical examples for the above.

- 5- Use a family tree to interpret probable Mendelian patterns of inheritance.
- 6- Calculate the probability to be affected with or carry a single-gene diseases (autosomal dominant, autosomal recessive and X-linked inheritance)
- 7-Clinical examples of each.

Definition

Alleles

Homozygous

Heterozygous

- **Allele**

Alternative form of a genetic locus, a single allele for each locus is inherited from each parent

- **Homozygous :**

An organism that has two identical alleles of a gene.

- **Heterozygosity:**

- The presence of different alleles at one or more loci on homologous chromosomes.

- **Phenotype:**

What is actually observed physically or clinically

- **Genotype:**

Individuals genetic constitution at a locus

- **Congenital:**

A condition is present at birth.

Alleles

different forms of the same gene





Homozygous

BB

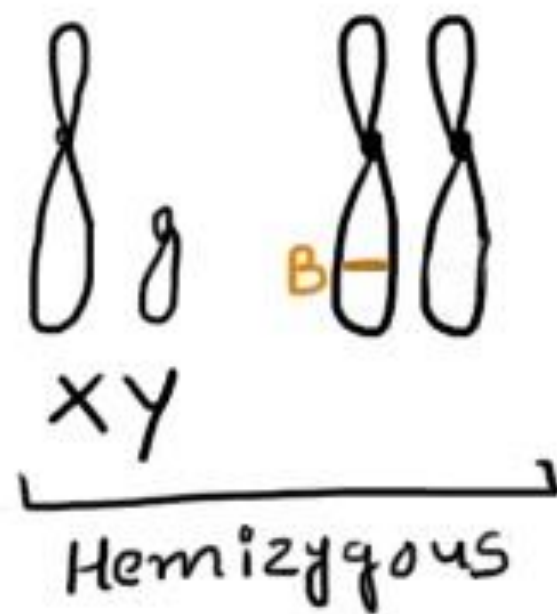
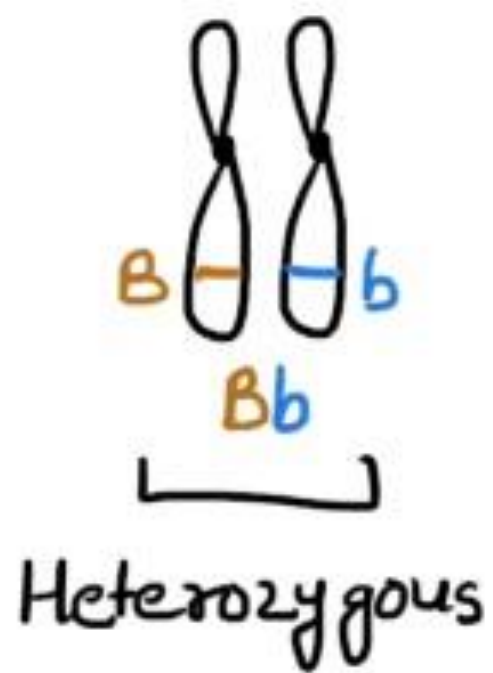
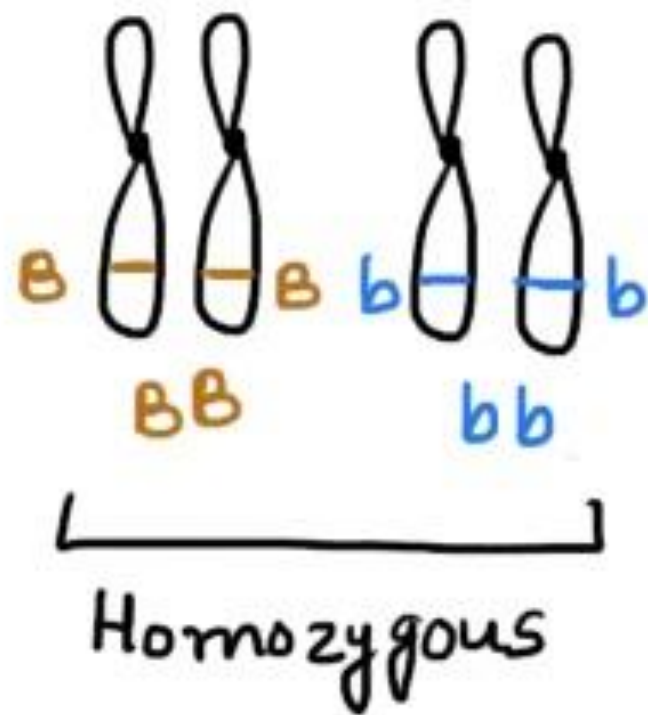


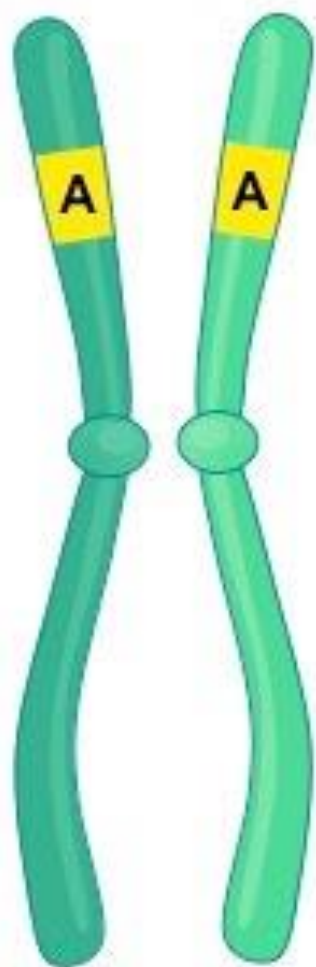
Heterozygous



Homozygous

bb





Homozygous
Alleles are same

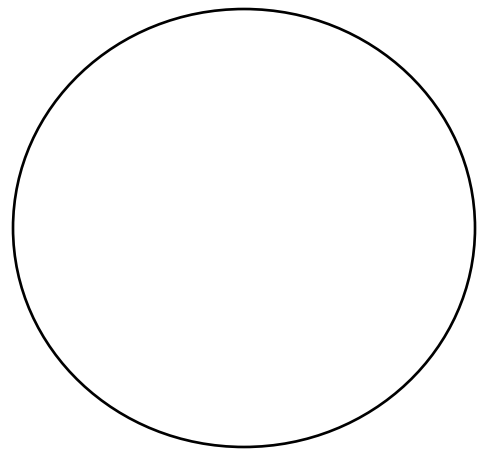


Heterozygous
Alleles are different

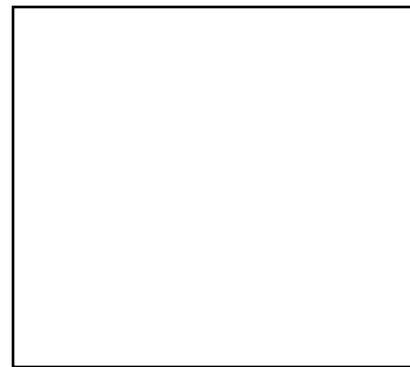


Hemizygous
*Only one allele
(e.g. XY)*

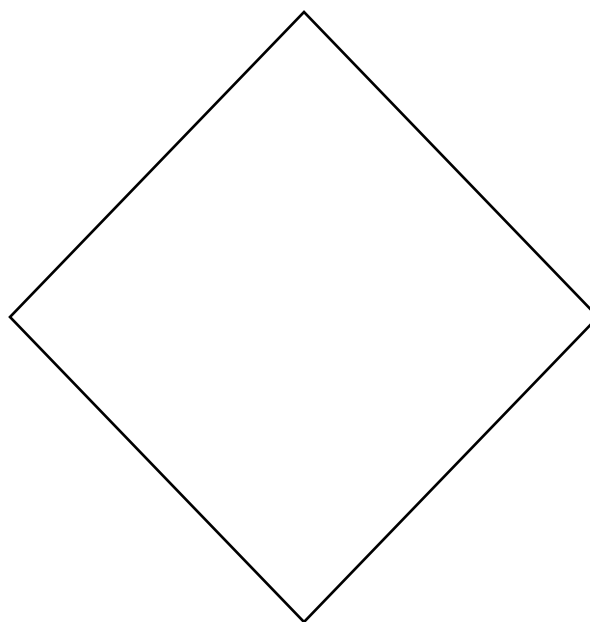
Draw a pedigree



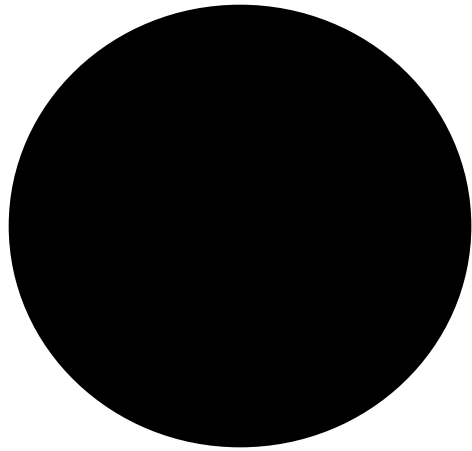
Normal female



Normal male



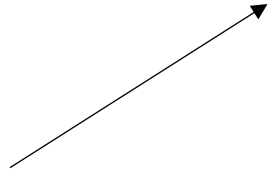
Unspecified Sex



Affected female



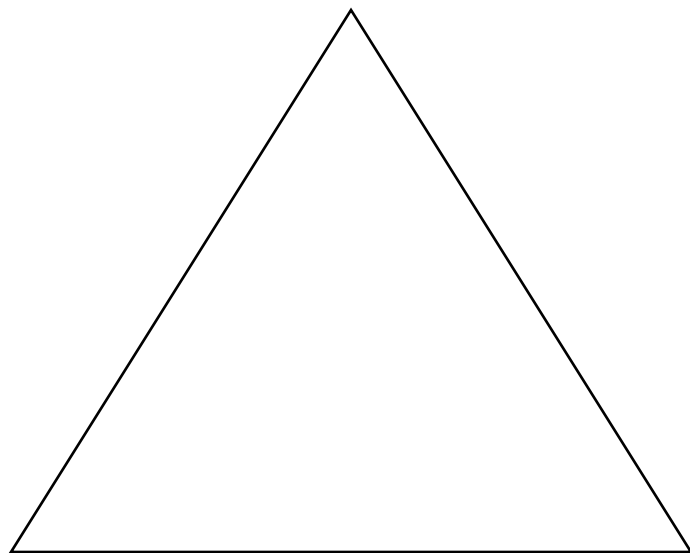
Affected male



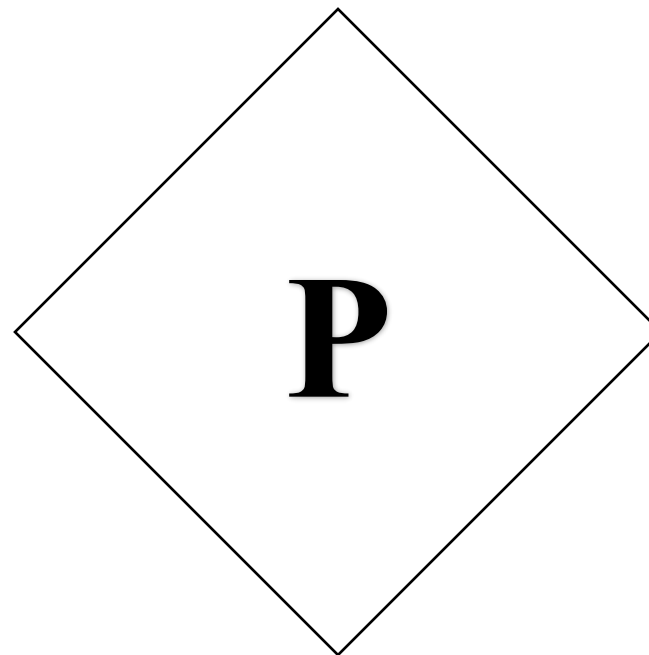
Proband

Index case

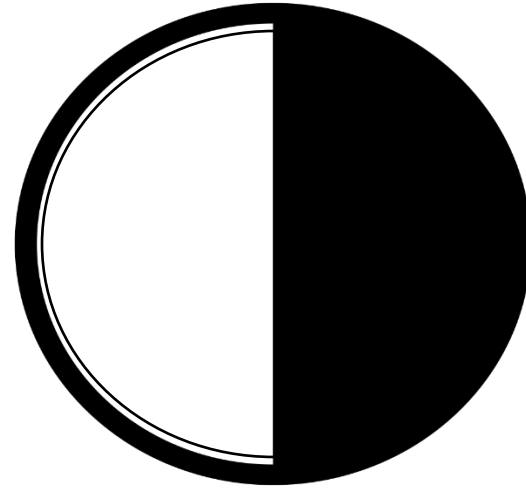
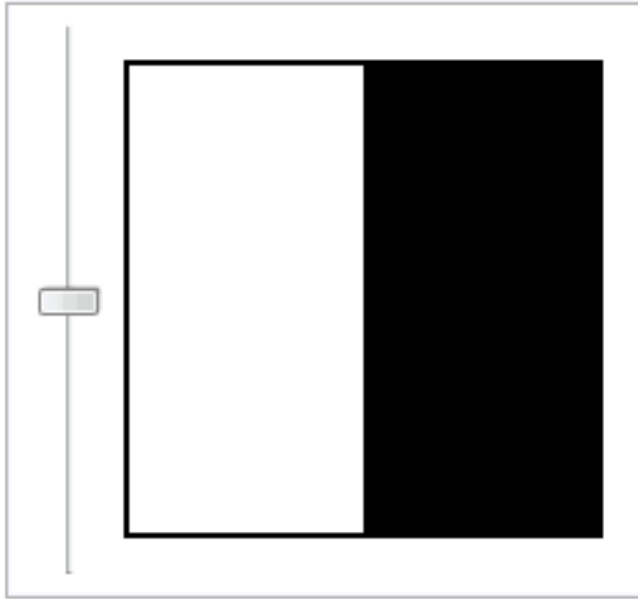
First person examined in the family



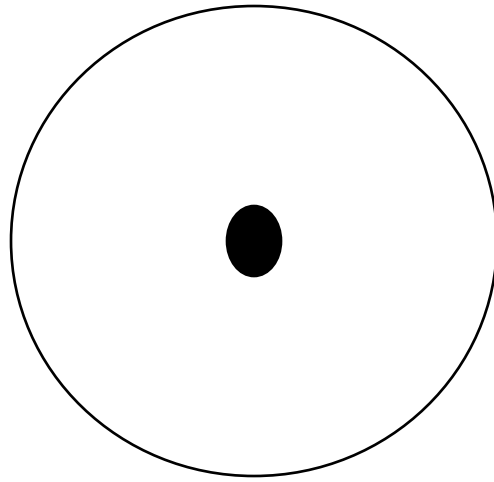
Abortion



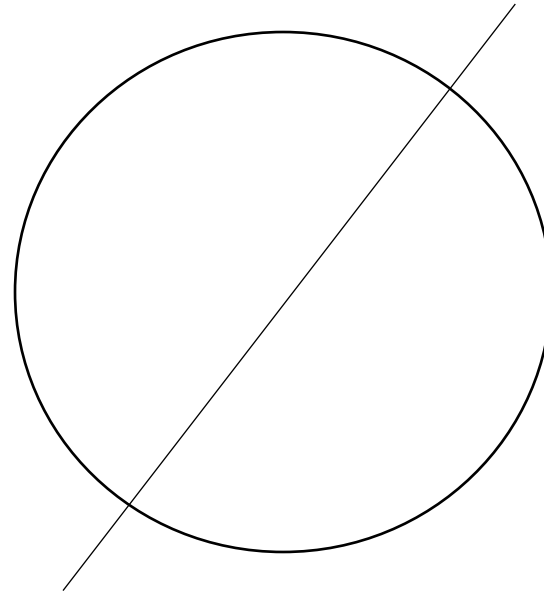
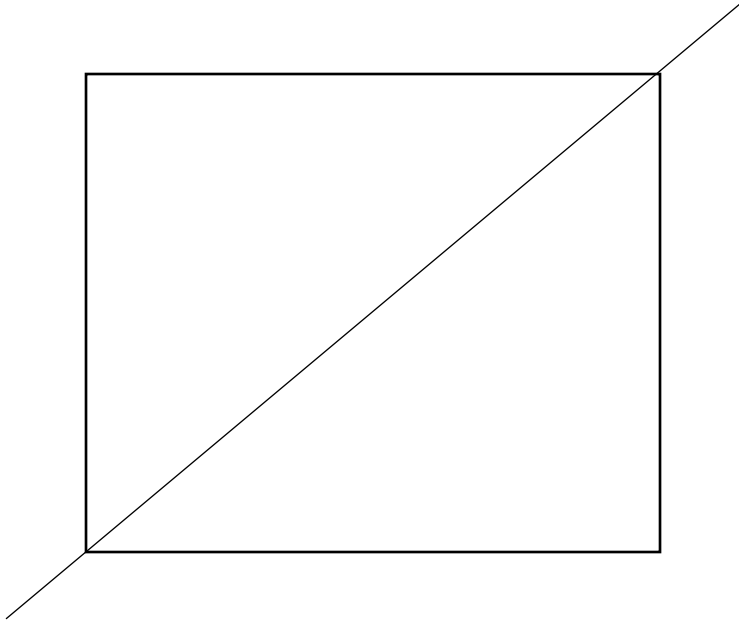
Pregnancy



Carrier autosome

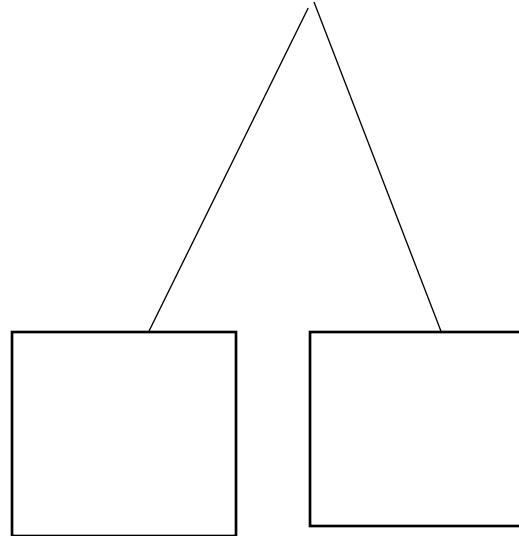


Carrier X-Linked

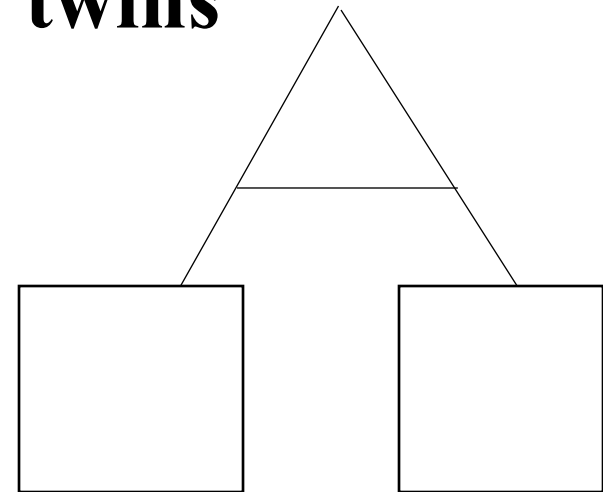


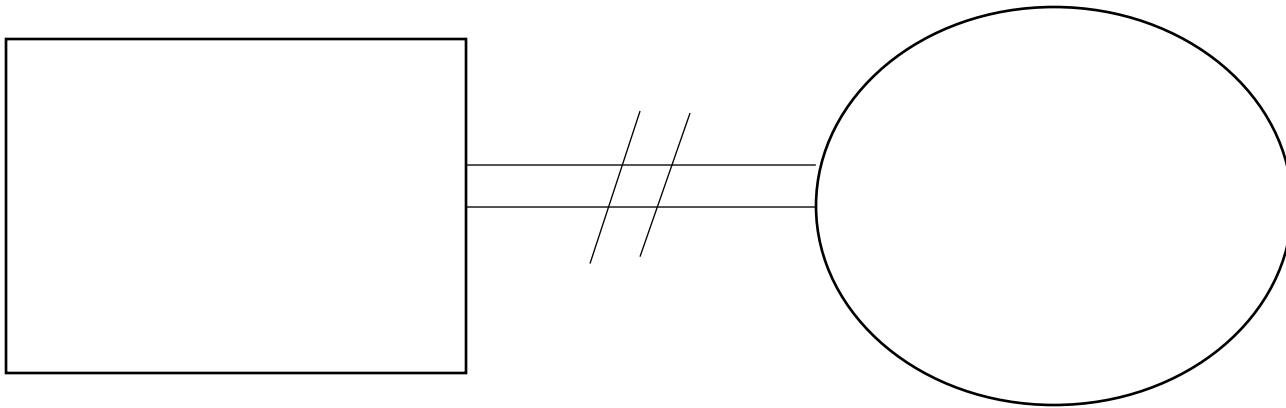
Deceased

Dizygotic twins



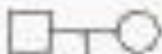
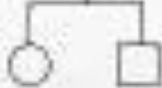
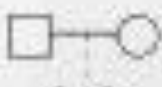
Monozygotic twins





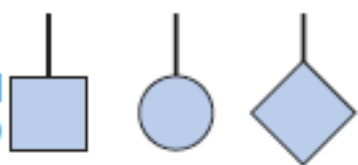
**Married
Consanguineous
Separation**

Standardized Pedigree Symbols and Relationships

	Male		Couple (horizontal line connects mates)
	Female		Offspring (vertical line connects parents with offspring)
	Sex unspecified		Adopted in
	Proband		Adopted out
	Affected		Monozygotic twins
	Carrier (autosomal)		Dizygotic twins
	Carrier (X-linked)		Zygosity unknown
	Deceased		
	Divorced		
	Consanguinous mating		

Individuals

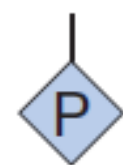
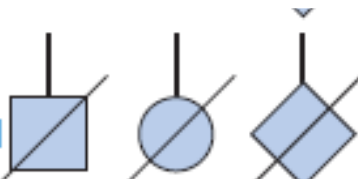
Normal
(male, female, unknown sex)



Affected individual

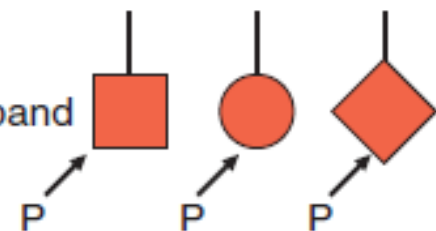


Deceased individual

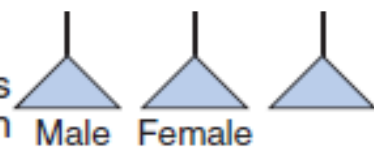


20 wk

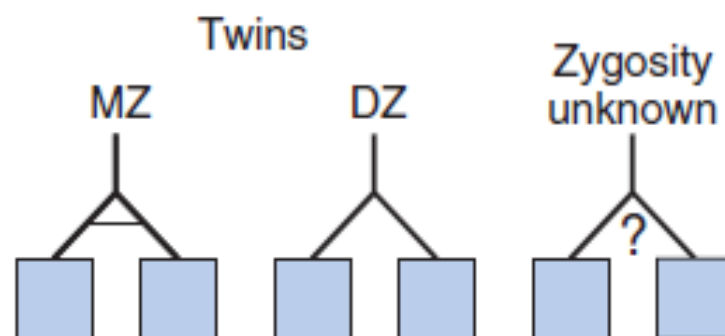
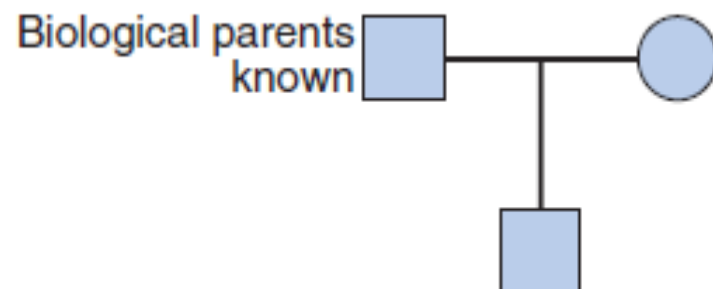
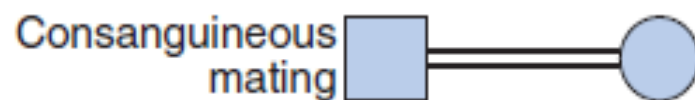
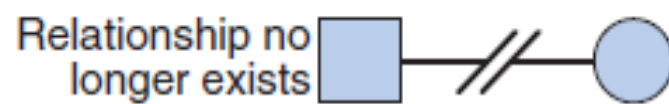
Proband

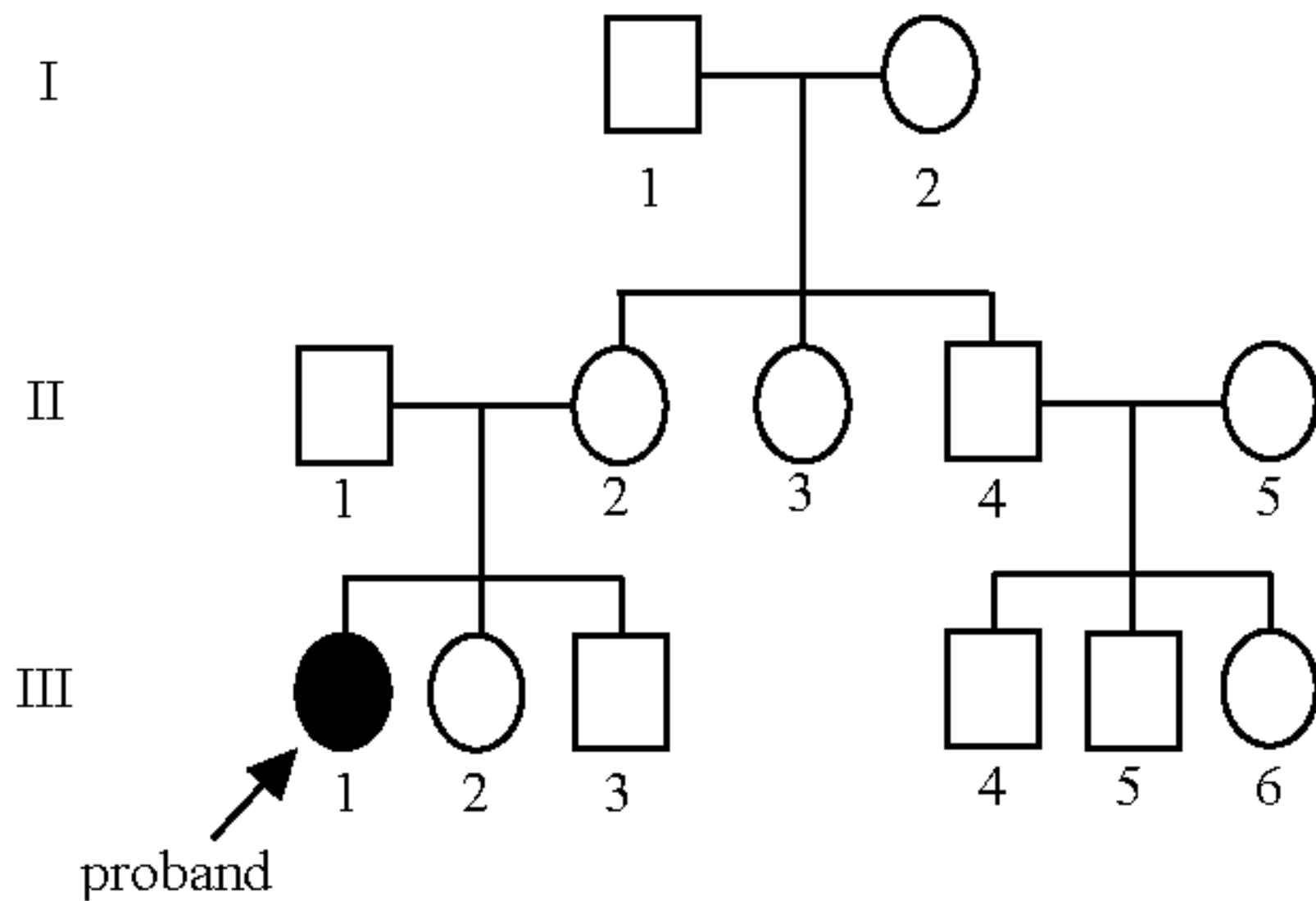


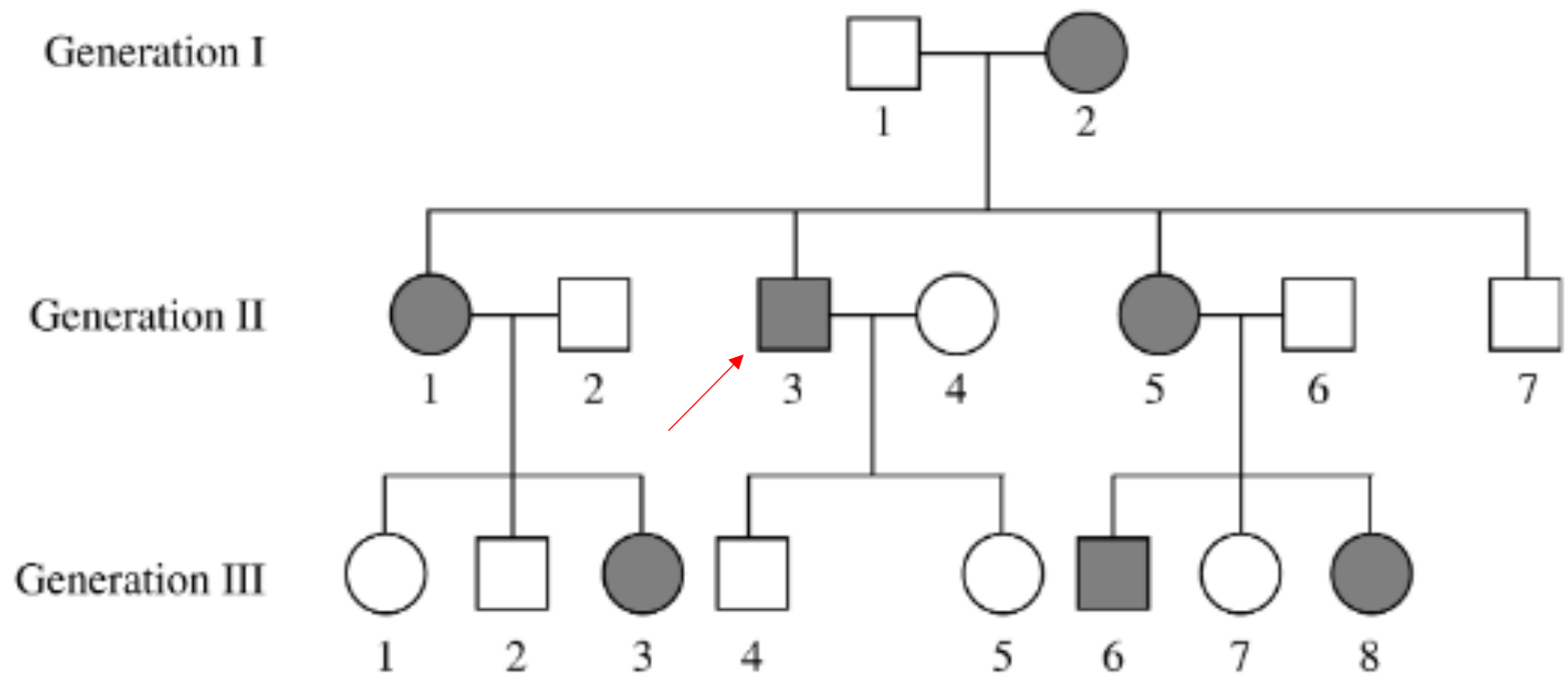
Spontaneous
abortion



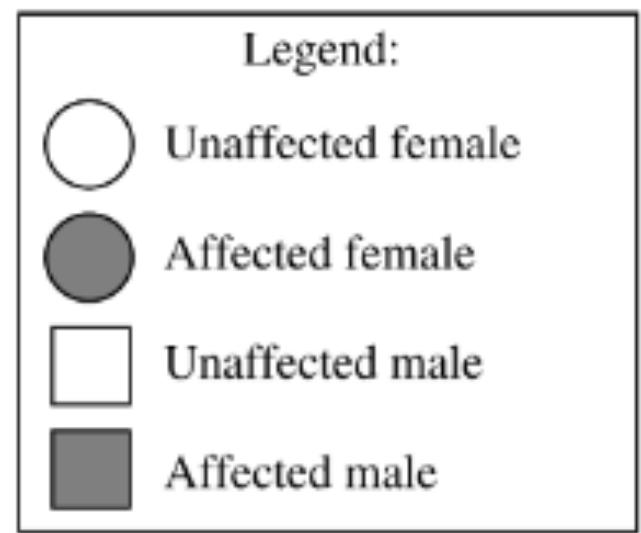
Relationships







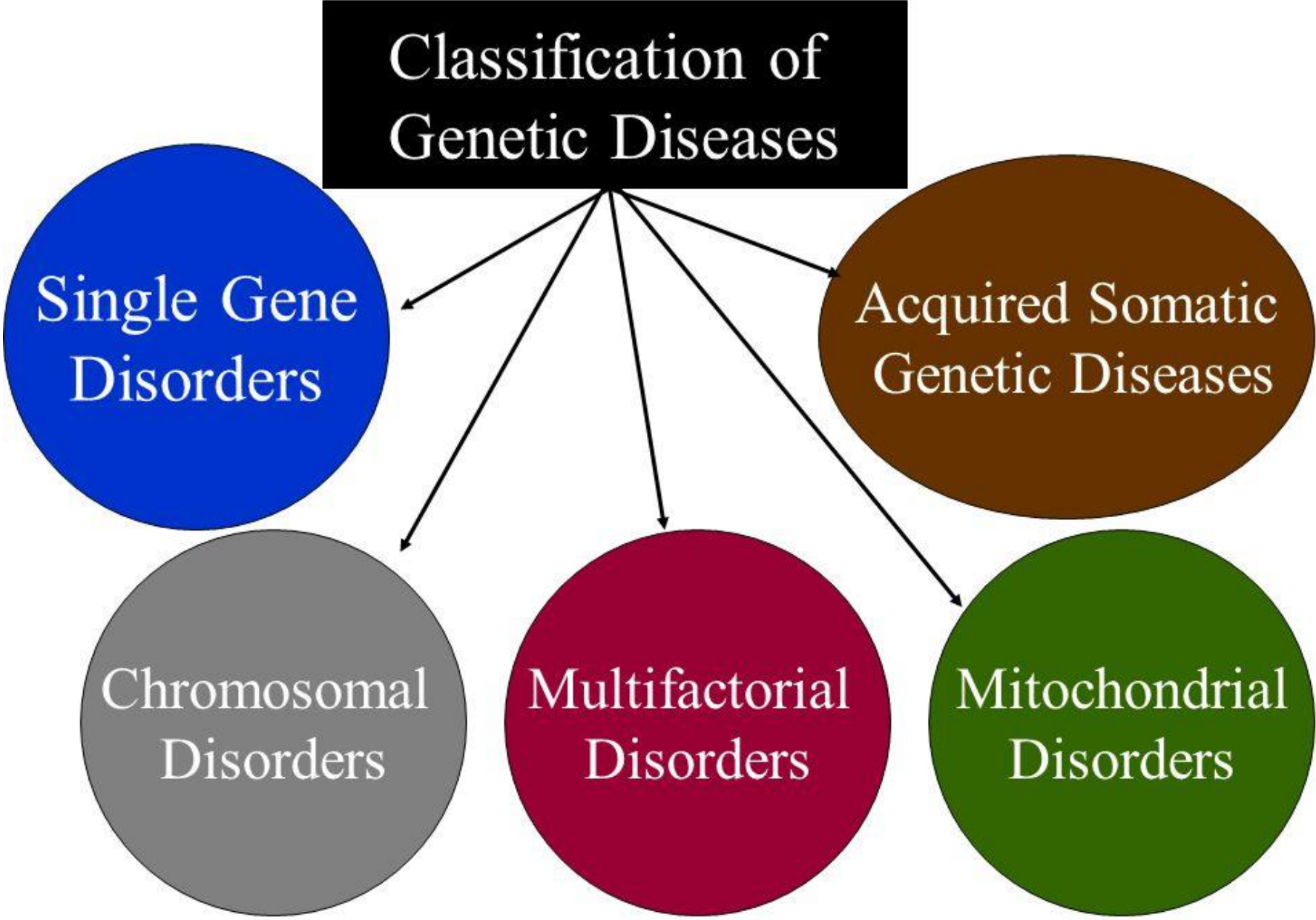
**Second generation
individual number 3**



- **Ahmad and Fatima are related by blood and married since seven years ago. They brought their son Khaled to the genetic clinic he has a genetic condition, Khaled has 3 sisters two of them having the same genetic condition . Fatima is pregnant. Family history revealed that Ahmad's parents were deceased.**
- **Draw a Three Generation Pedigree for this family.**

- **Mohamed and Laila came to the genetic clinic seeking for advice regarding their daughter Fatima, she has a genetic condition. Fatima has 2 brothers one of them has the same genetic condition. Family history revealed that both Mohammed and his sister have a mild manifestation of such disorder. The deceased paternal grandmother was similarly affected. Laila is pregnant now.**
- **Draw a Three Generation Pedigree for this family.**

Classification of Genetic Diseases



```
graph TD; A[Classification of Genetic Diseases] --> B((Single Gene Disorders)); A --> C((Acquired Somatic Genetic Diseases)); A --> D((Chromosomal Disorders)); A --> E((Multifactorial Disorders)); A --> F((Mitochondrial Disorders));
```

Single Gene Disorders

Acquired Somatic Genetic Diseases

Chromosomal Disorders

Multifactorial Disorders

Mitochondrial Disorders

Single Gene Disorders

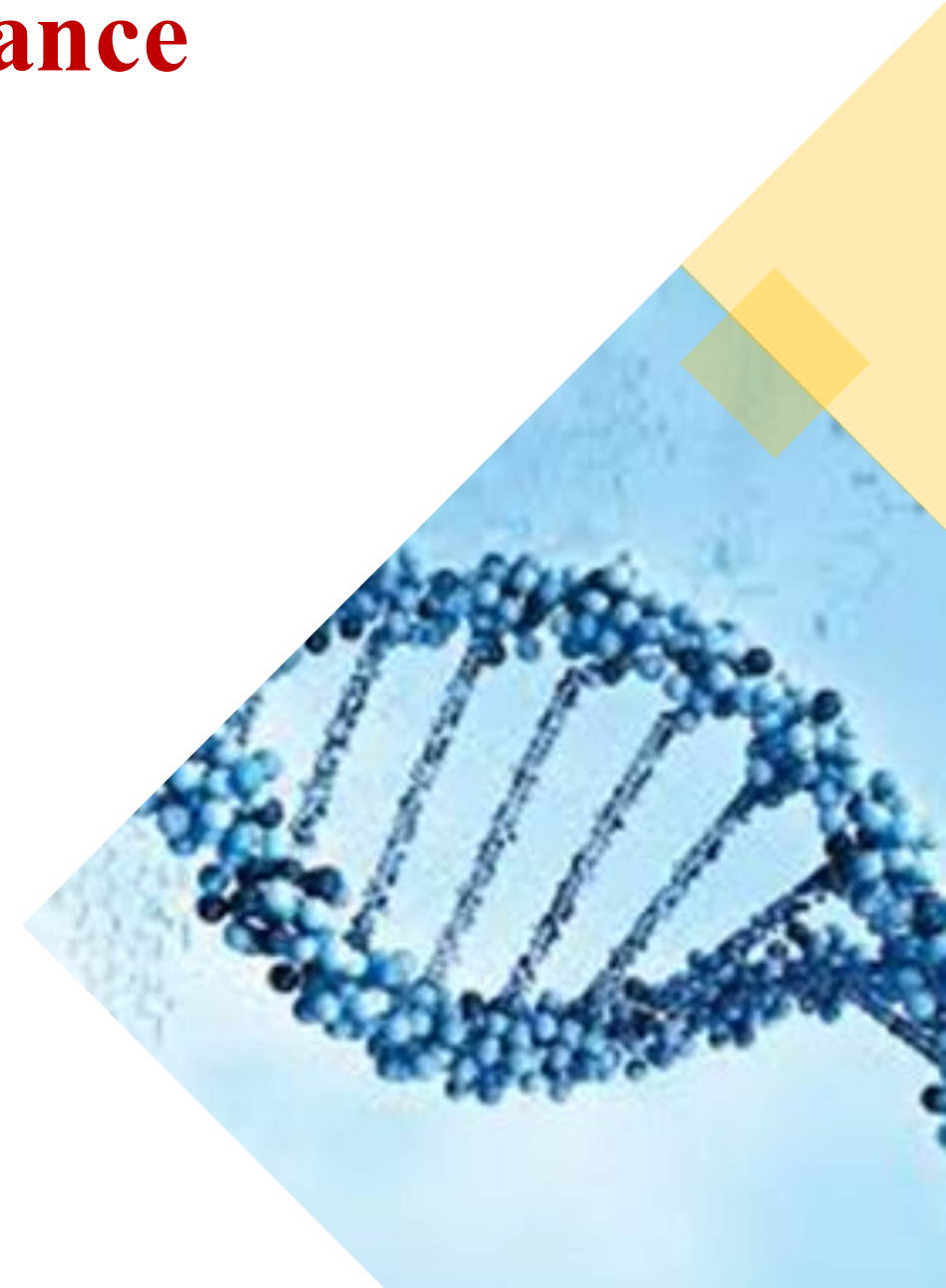


Patterns of Mendelian inheritance

- Autosomal dominant
- Autosomal recessive
- Sex-linked (X-linked recessive, X-linked dominant, Y-linked).

Mendelian Inheritance

- A trait or disorder that is determined by a gene on an autosome is said to show autosomal inheritance, whereas a trait or disorder determined by a gene on one of the sex chromosomes is said to show sex-linked inheritance.





Autosomal Inheritance



Autosomal Dominant Inheritance

Disorder



Autosomal Dominance Inheritance

Key

D = autosomal dominant disorder

d = healthy allele

Healthy



There are four possible allele combinations that their children may inherit



Dd



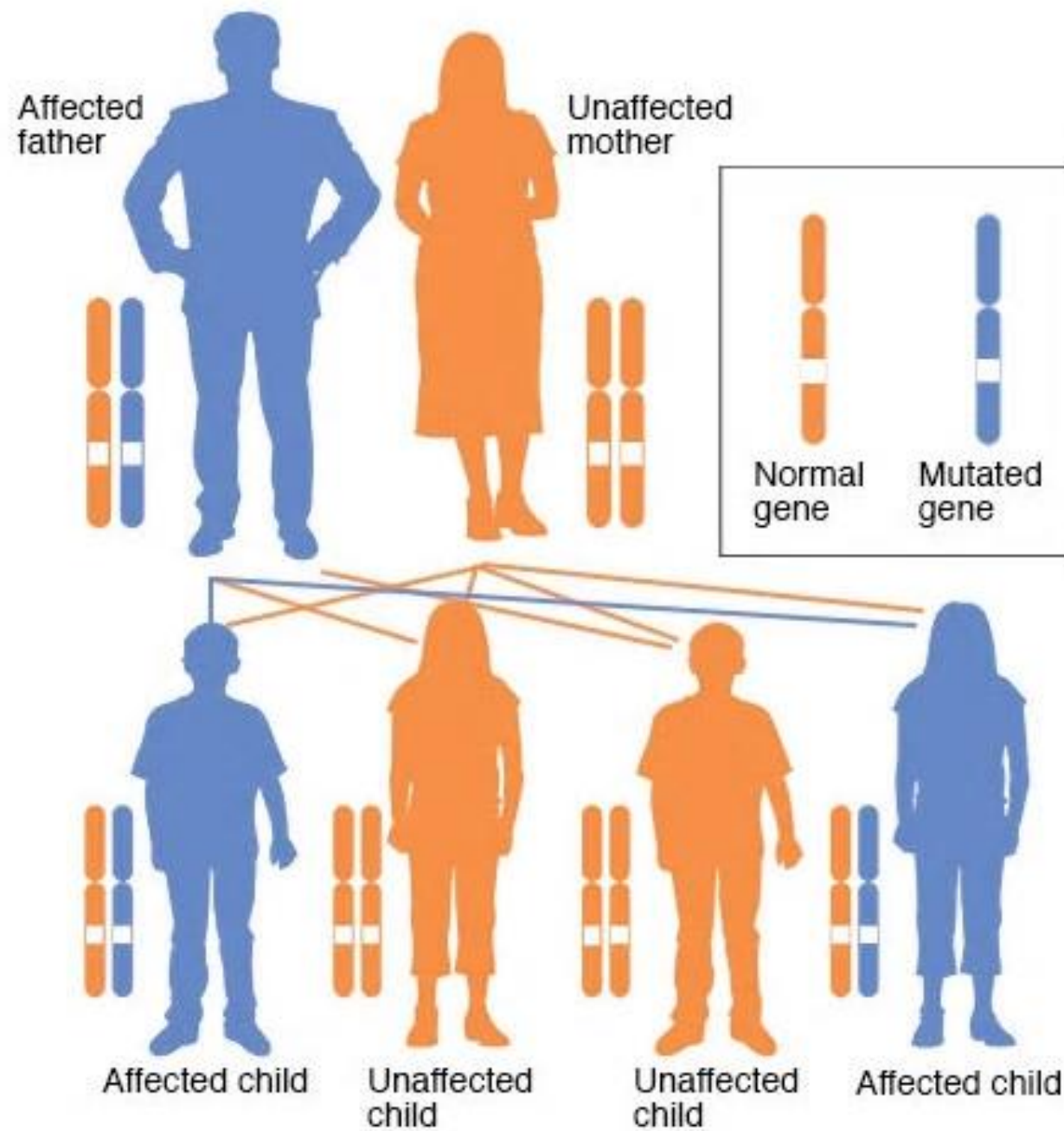
dd



Dd

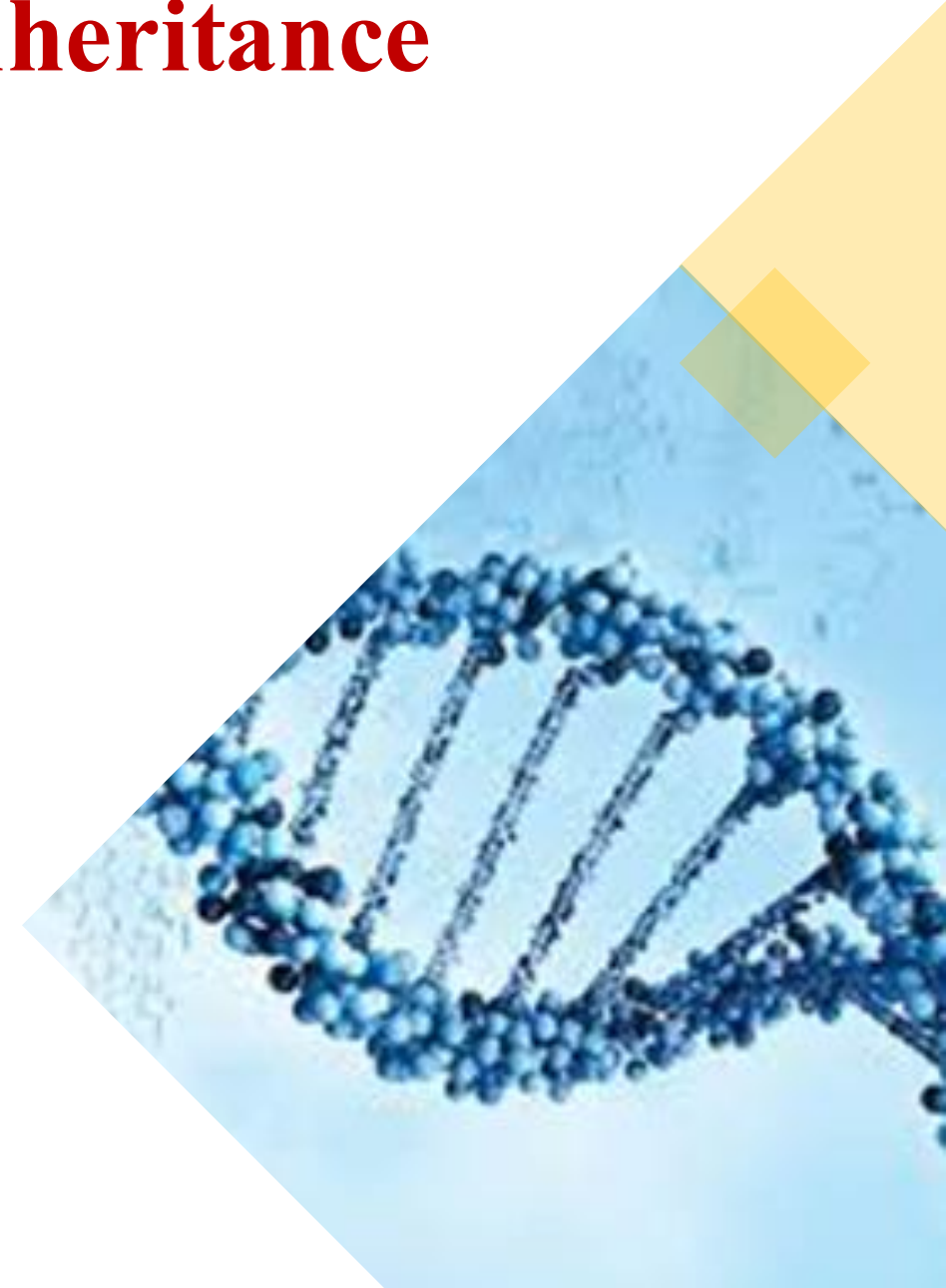


dd



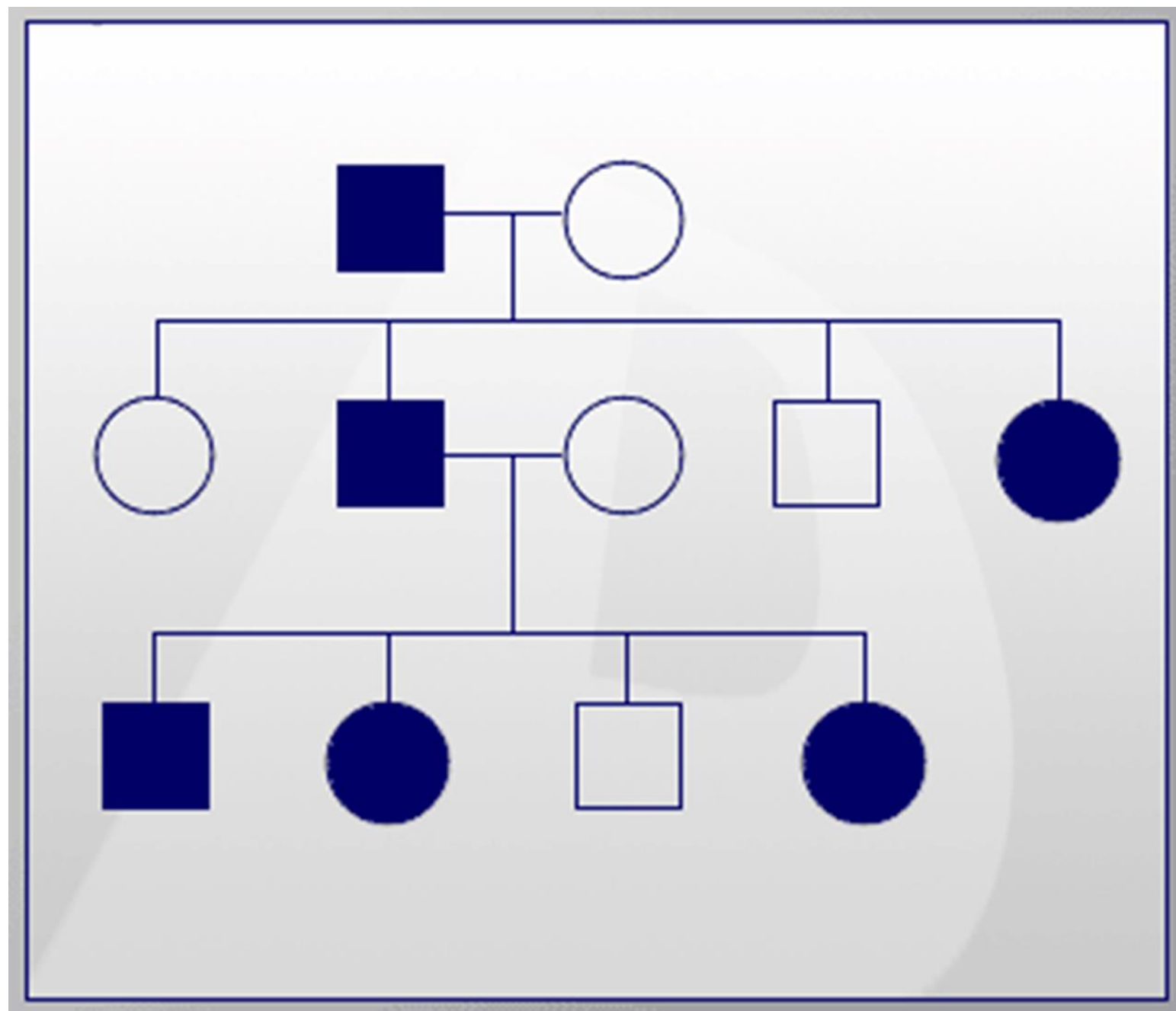
Autosomal Dominant Inheritance

- One parent must be affected
- Can manifest in a heterozygote state
- Usually transmitted from one generation to the next without skipping



- Both males and females are equally affected
- All forms of transmission between the sexes are observed
- Male to male transmission is a key feature

- With each pregnancy, there is a one in two (50%) chance the offspring will inherit the disease allele.
- Autosomal dominant inheritance is often called vertical inheritance because of the transmission from parent to offspring.



Autosomal Dominant Inheritance

Recurrence risk is

50% per each pregnancy

Examples

But

AD traits can show:

- Pleiotropy
- Variable expressivity
- Reduced penetrance
- New mutations

Pleiotropy

- A single gene that may give rise to two or more apparently unrelated effects

So

It's common in AD disorders to manifest in different systems of the body in a variety of ways.

Tuberous sclerosis

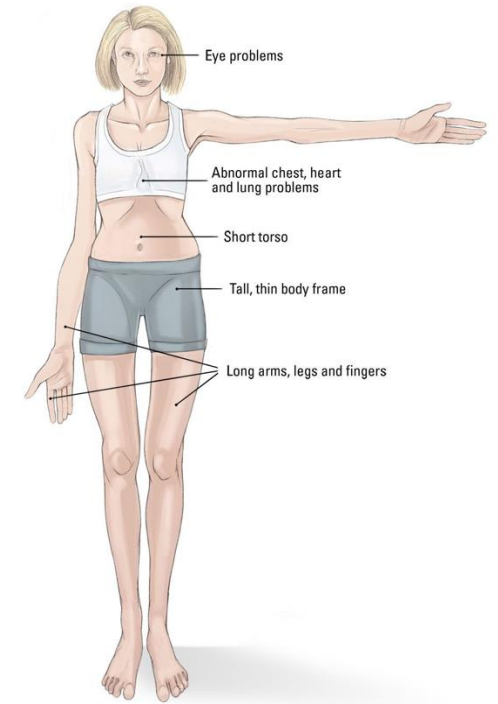
- Growth of numerous noncancerous (benign) tumors in many parts of the body.
- Learning difficulties
- Epilepsy
- Facial rash(adenoma sebaceum)



Marfan syndrome

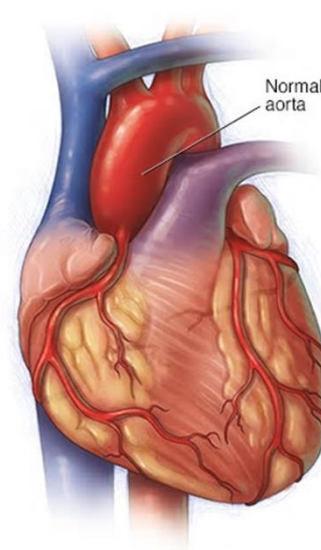
- Its characterized by defect in three major systems

Ocular, cardiovascular, skeletal

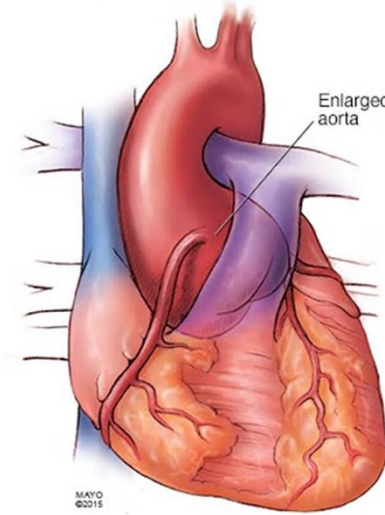




**Thumb
sign**



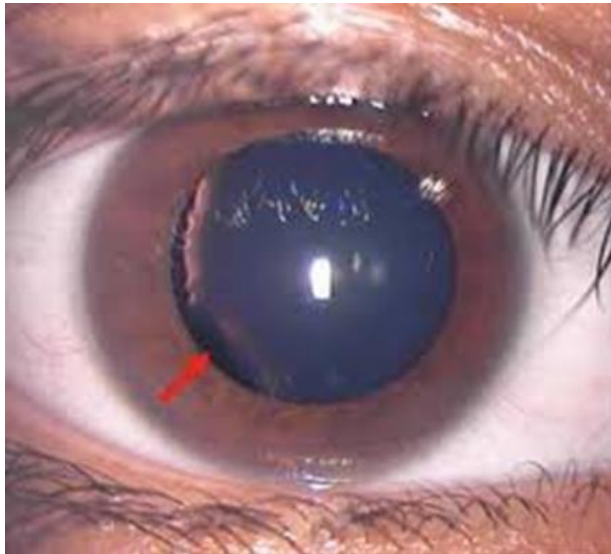
Normal heart and aorta



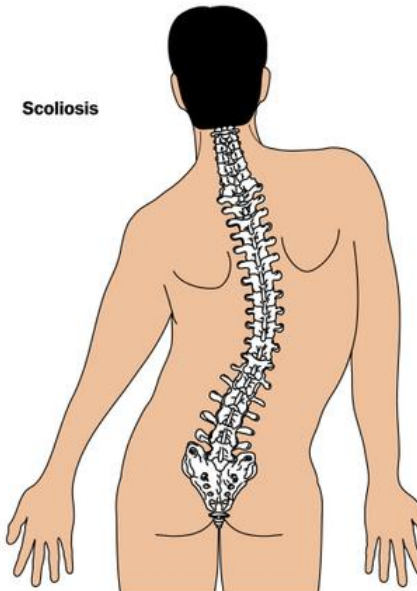
Heart with an enlarged aorta

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**Aortic
dilatation**



Ectopia lentis



Scoliosis

Variable Expressivity

(Degree of severity of the disease phenotype)

The clinical features in AD disorders can show **striking variation** from person to person, even in the same family.

AD Polycystic Kidney Disease

- Some individuals develop renal failure in early adulthood
- Others have just a few renal cysts that do not affect renal function.

Neurofibromatosis

A disease with highly variable expression

- Some patient have only Café-au-lait spots
- Some patients have only (Café-au-lait spots + iris Lisch nodules + few neurofibromatosis)
- Other patients have a much more severe expressions.

Neurofibromatosis



Reduced or Incomplete Penetrance

- When a person has a disease-causing gene and not manifesting the disease phenotype at all
- This person transmits the disease-causing gene to his children who become affected.
- This person always has an affected parent or relatives.

Retinoblastoma



Treacher Collins Syndrome



Deformities of the ears, eyes, cheekbones, and chin(facial bone)
very small jaw and chin (micrognathia). Other features may
include cleft palate and hearing loss

New Mutation

- In AD disorders an affected person usually has an affected parent.

But

- Sometime AD trait appears in individuals with no family history of the disorders

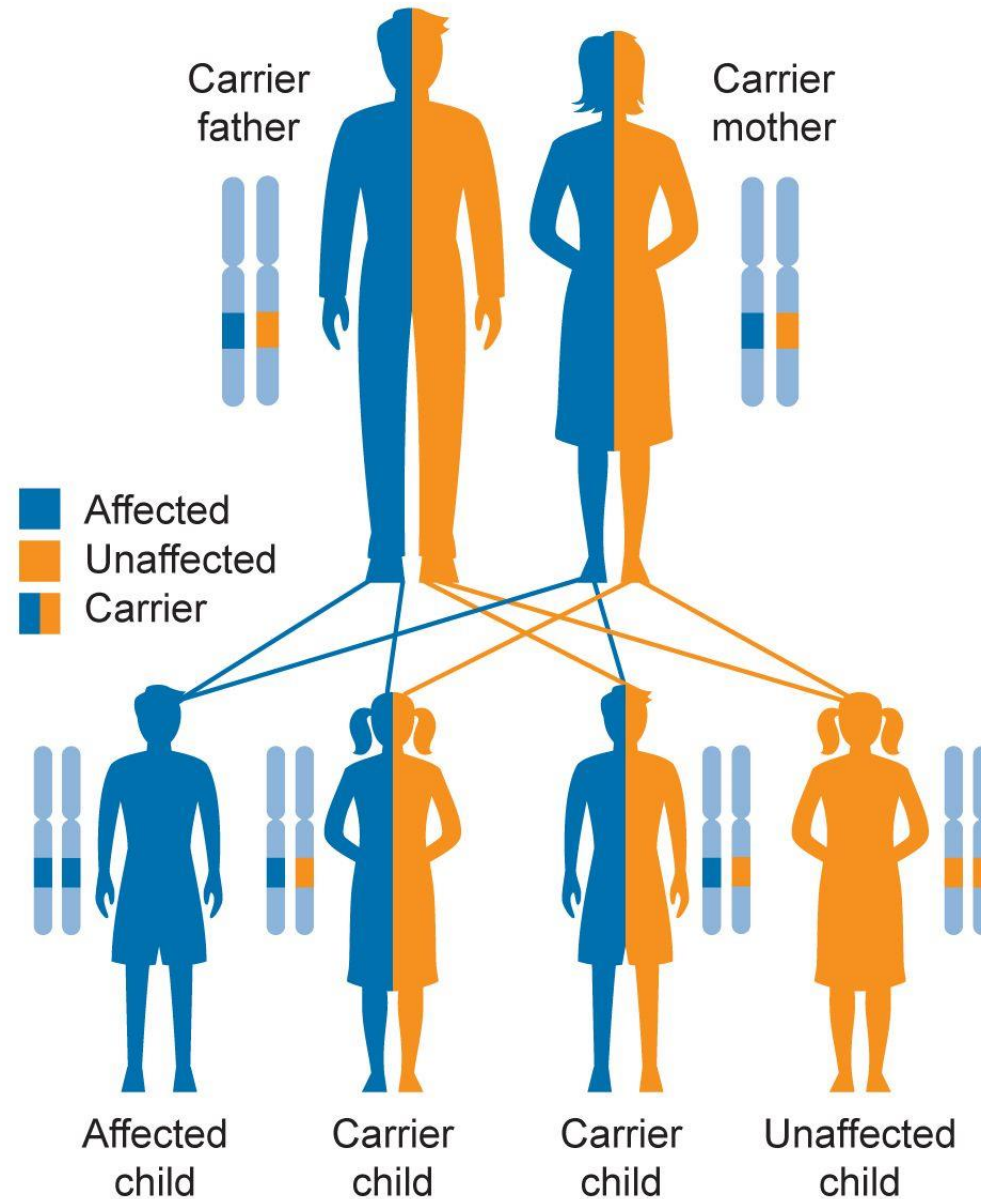
- **Example:** Achondroplasia (short limbed dwarfism), in which parents have normal stature.





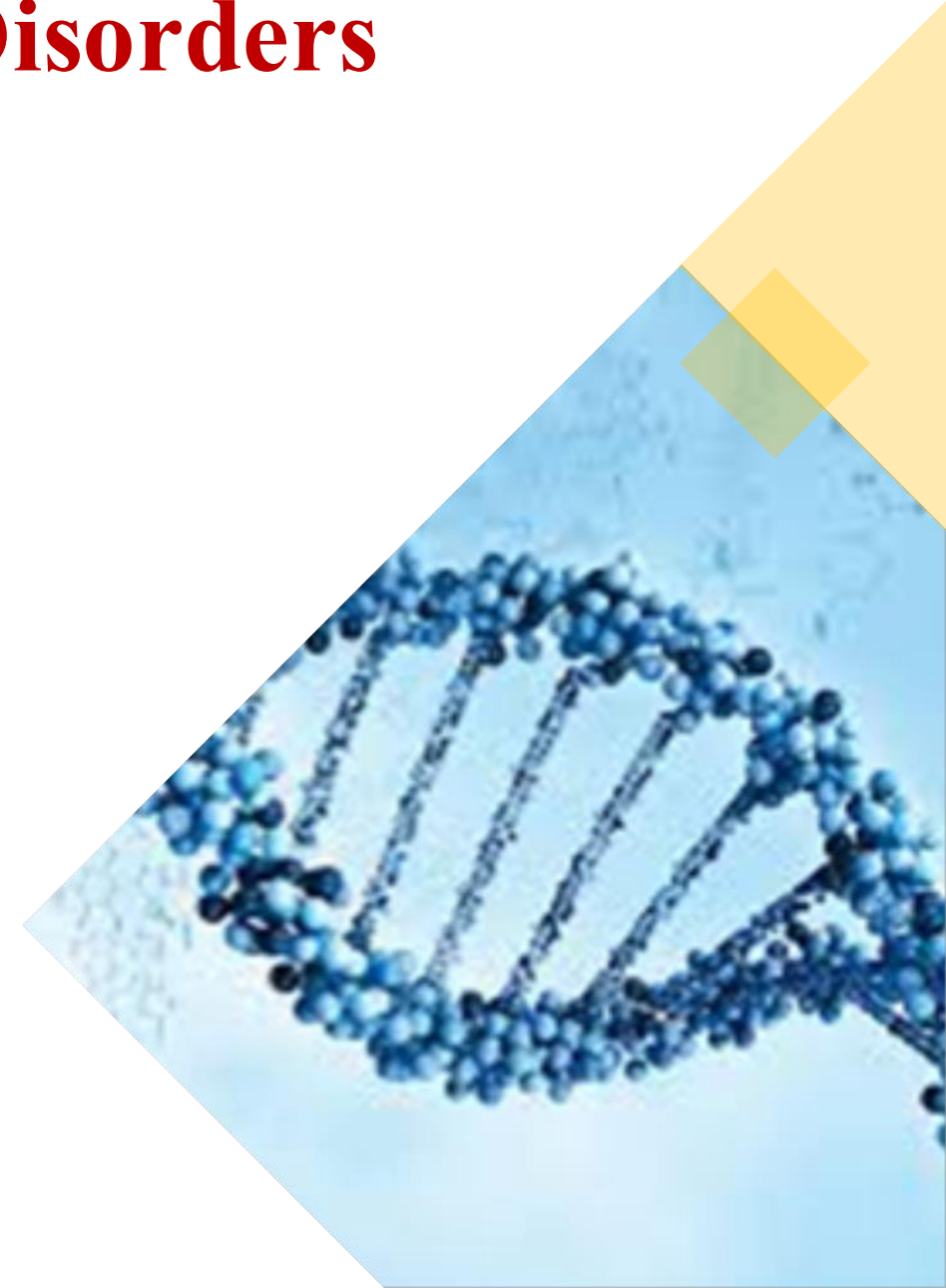
Autosomal Recessive Inheritance

Autosomal Recessive Inheritance



Autosomal Recessive Disorders

- Parents are apparently normal
- Parents are heterozygous carriers
- Usually skipped generations

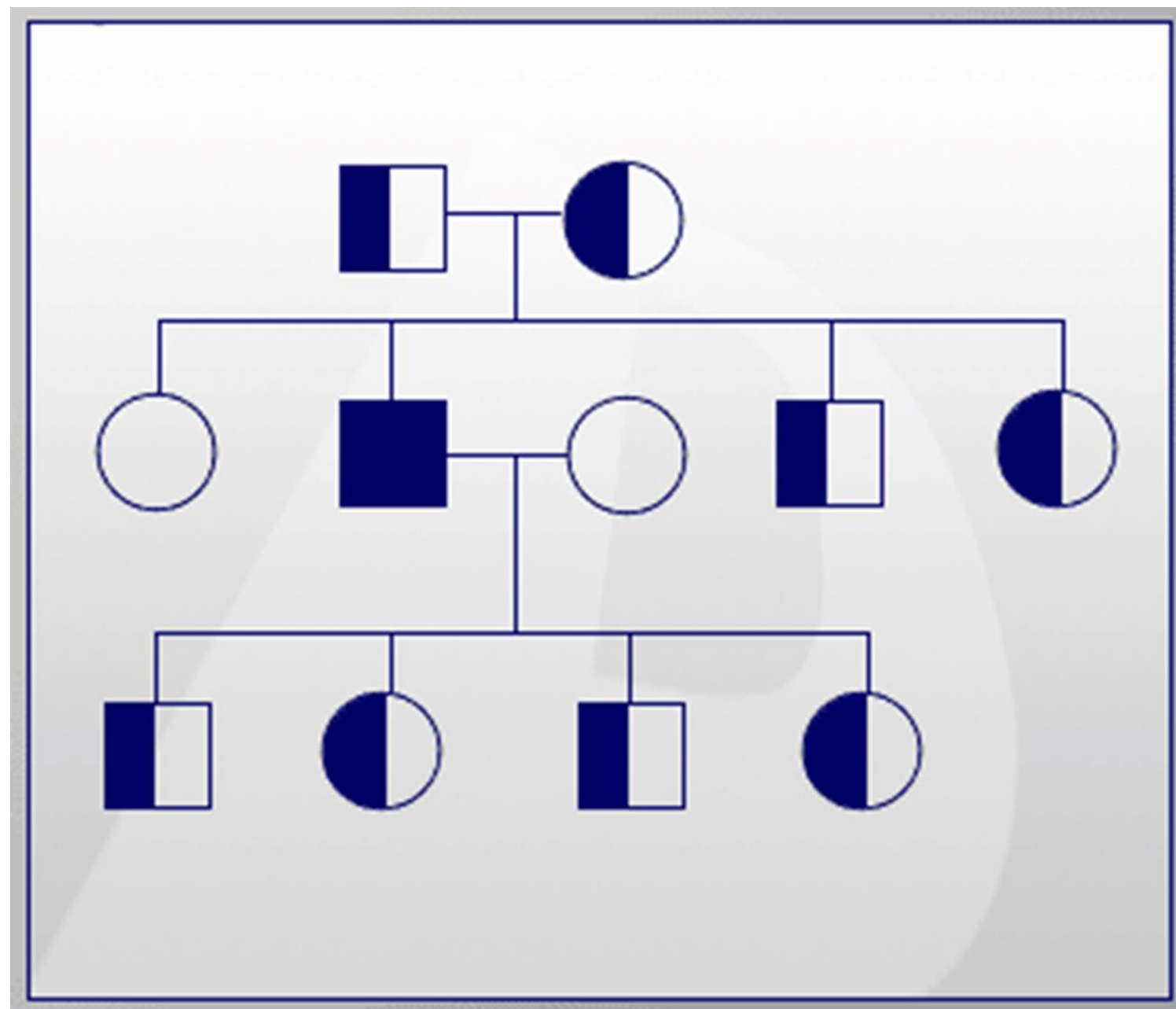


- Both males and females are equally affected
- Parents tend to be related more often than parents in general population.

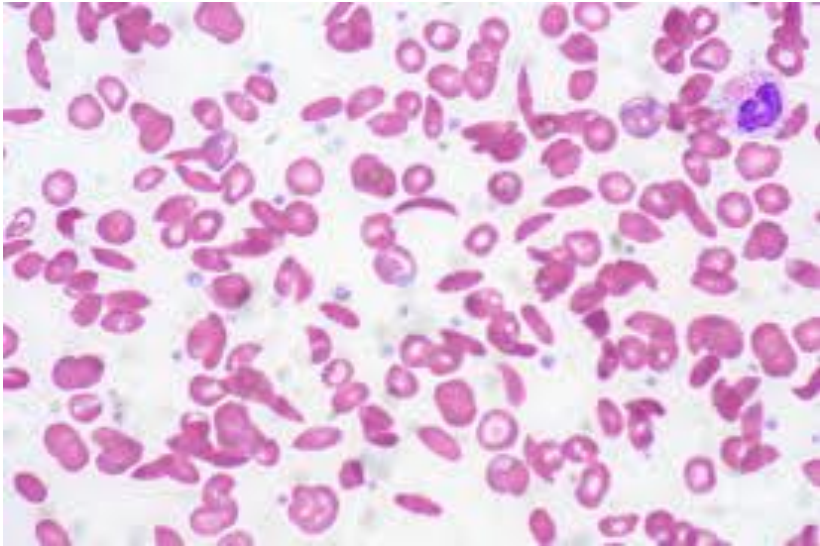
Autosomal Recessive Disorders

Recurrence risk is

25% per each pregnancy



Sickle Cell Anemia

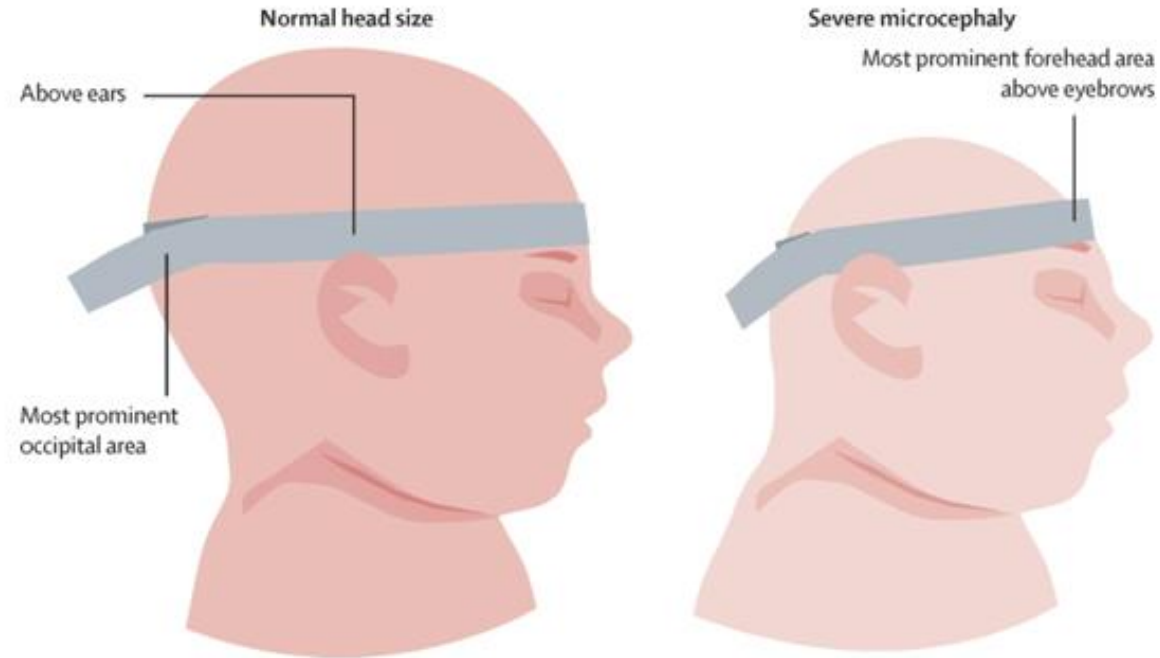


Normal Red Blood Cell



Sickle Cell

AR Microcephaly



Baby with typical head size



Baby with microcephaly



Baby with severe microcephaly





Albinism



Cystic Fibrosis

Cystic Fibrosis



Mucus blocks
air sacs (alveoli)
in the lungs



Mucus blocks
pancreatic ducts

Pancreatic
duct

Pancreas



Stomach

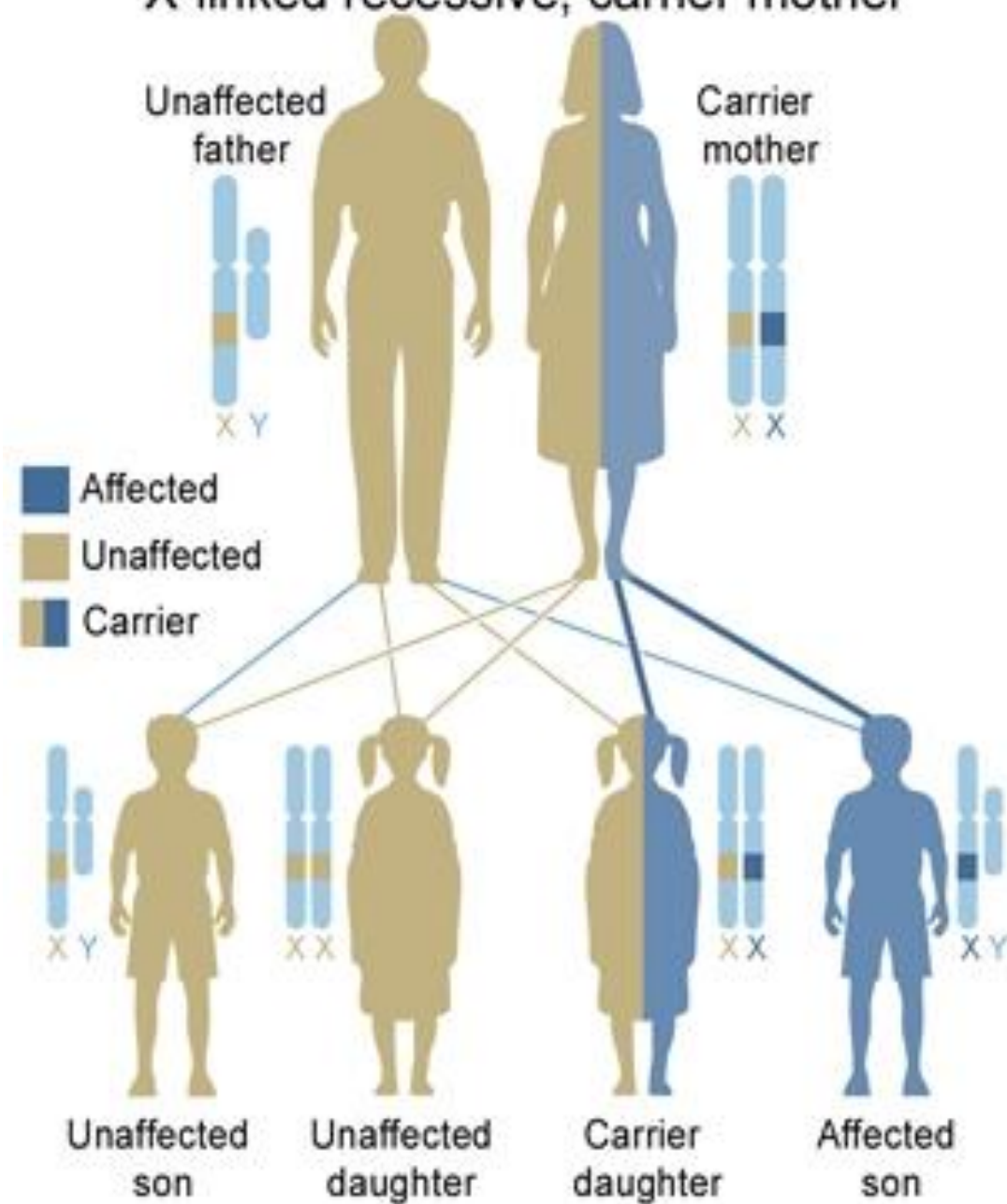


Sex Linked Inheritance



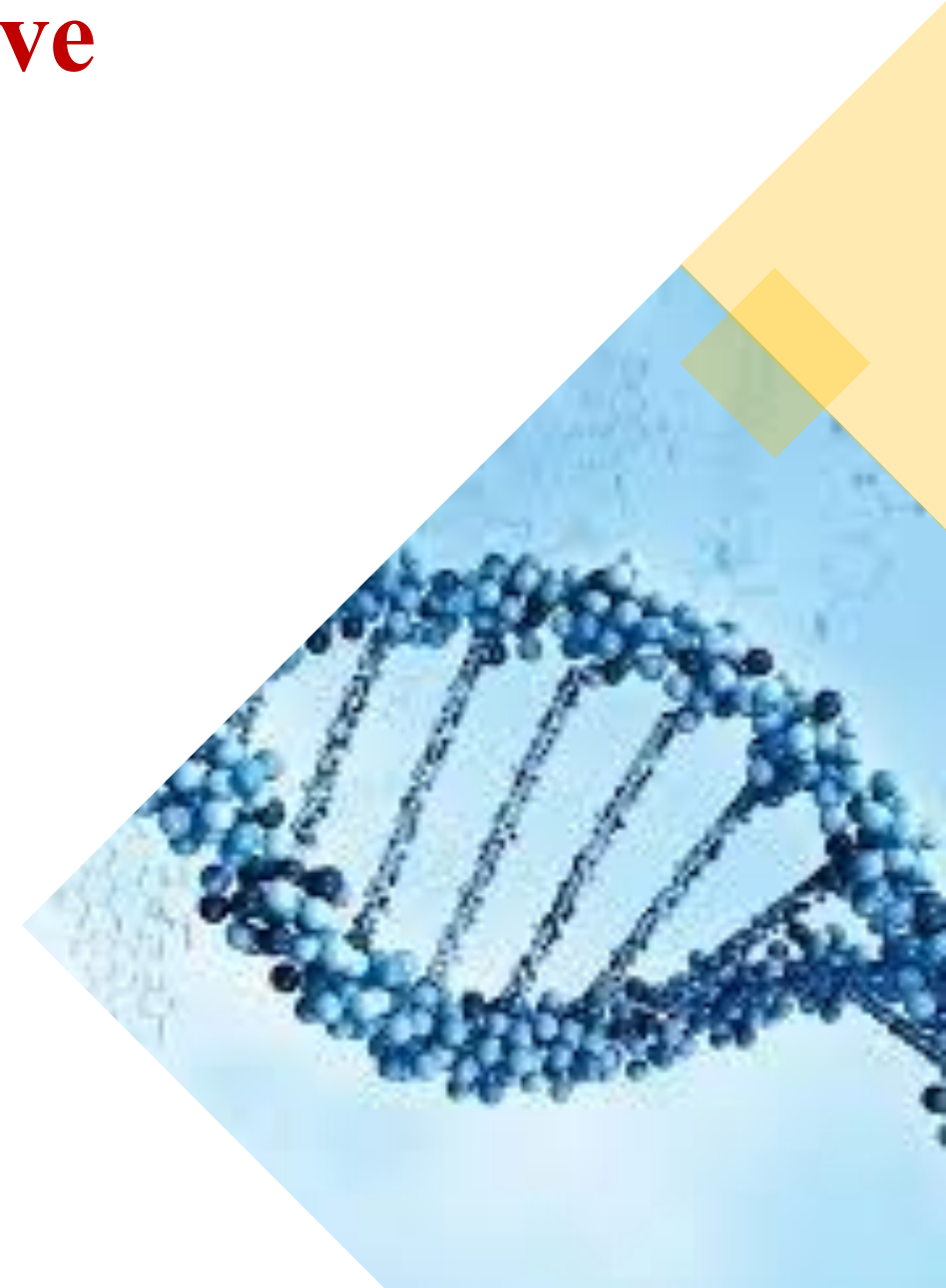
X-linked Recessive

X-linked recessive, carrier mother



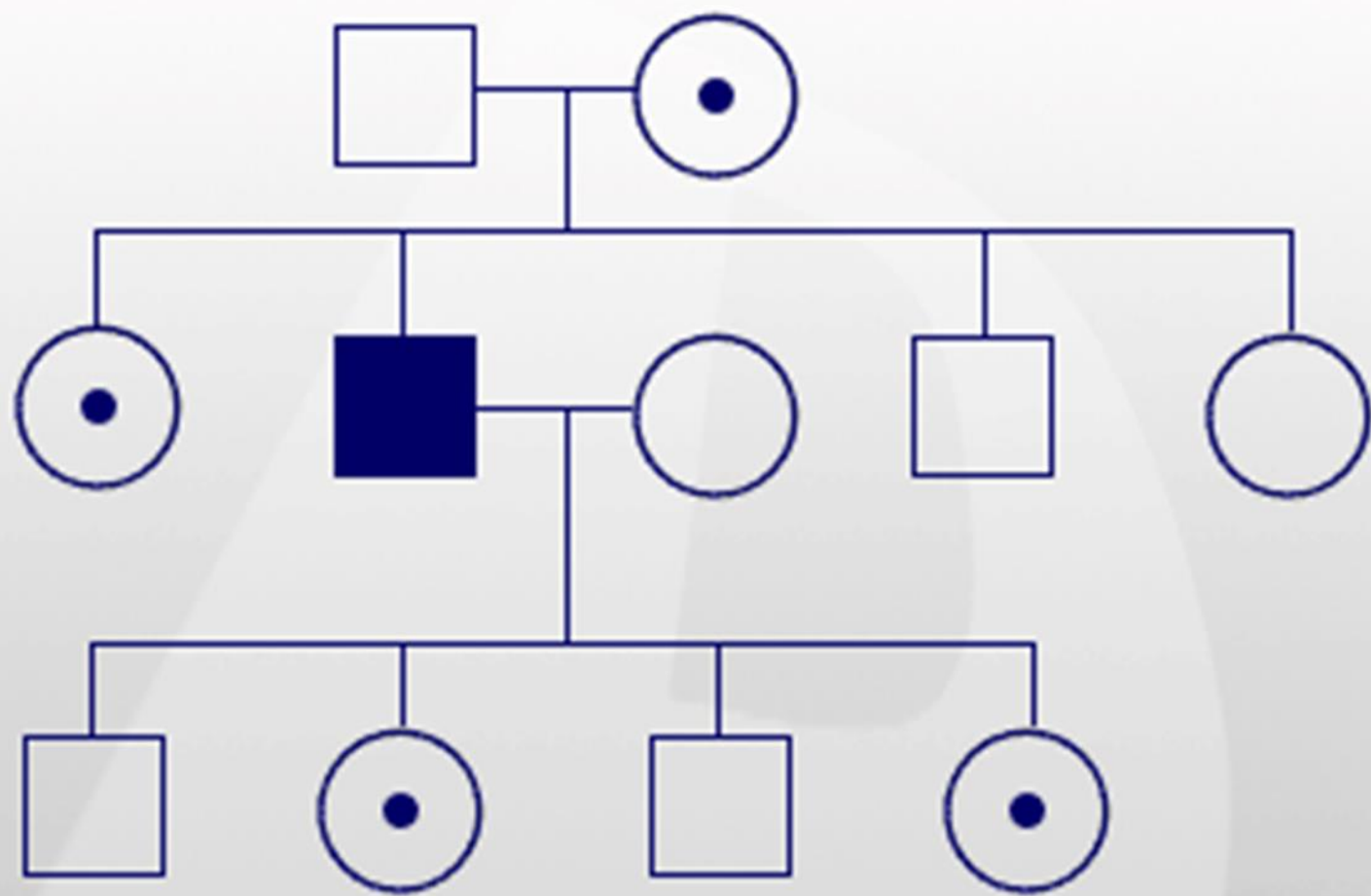
X-linked Recessive

- Transmitted through carrier females to affect their sons
- Affect males almost exclusively



- Since males are hemizygous for X-linked genes (they have only one X chromosome), any male with one copy of an X-linked recessive disease allele is affected

- For a carrier female, with each pregnancy there is a one in two (50%) chance her sons will inherit the disease allele and a one in two (50%) chance her daughters will be carriers.



Duchenne Muscular Dystrophy

- DMD is caused by changes (mutations) in the DMD gene. The DMD gene codes for the protein dystrophin.
- Dystrophin is mainly made in skeletal and heart muscle cells.

Other Examples of XLR

- Red green color blindness
- Hemophilia
- G6PD

Lyonization

The process of inactivation of one of X-chromosomes in females (Mary Lyon)

- The process of X-inactivation usually occurs randomly
- In embryogenesis after X-inactivation roughly
- A) In $\frac{1}{2}$ of the cells one of the X-chromosomes is active and the other is inactive
- B) In the other $\frac{1}{2}$ of the cells the other X-chromosome is the active one

- If X-chromosome inactivation is not random, allowing for the possibility that
- The active X-chromosome in most of the cells of a heterozygous female carriers bearing the mutant allele

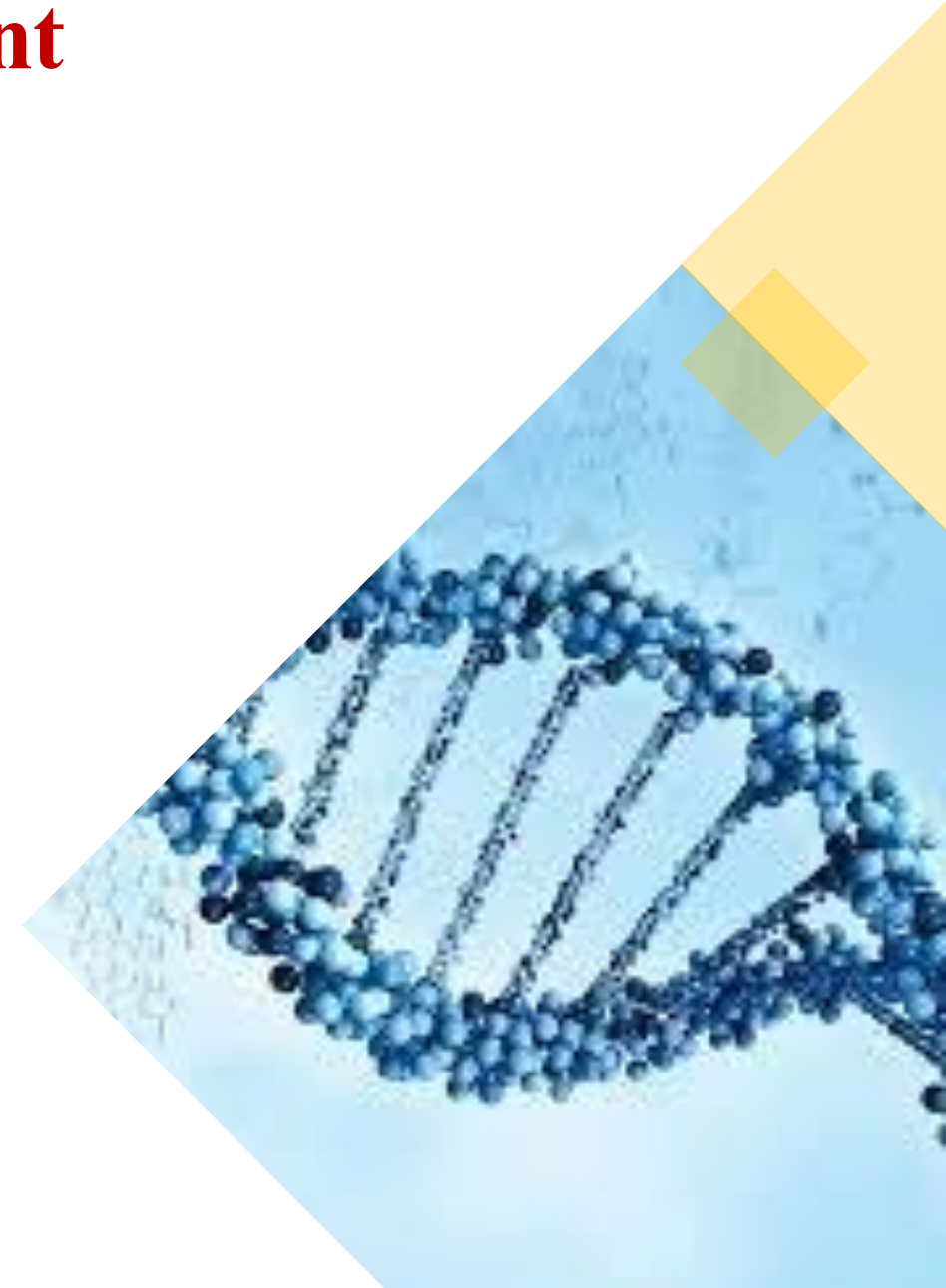
SO

The carrier female will exhibit some of symptoms and signs of the disease.

X-linked Dominant

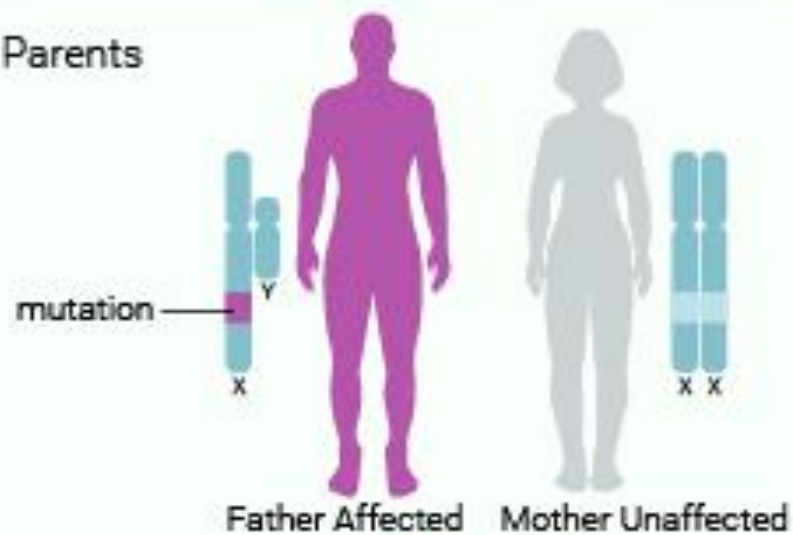
X-linked Dominant

- Uncommon
- Manifest in the heterozygous females as well as in the male who has the mutant allele on his single X-chromosome
- The affected male transmit the trait to all of his daughters and non of his sons

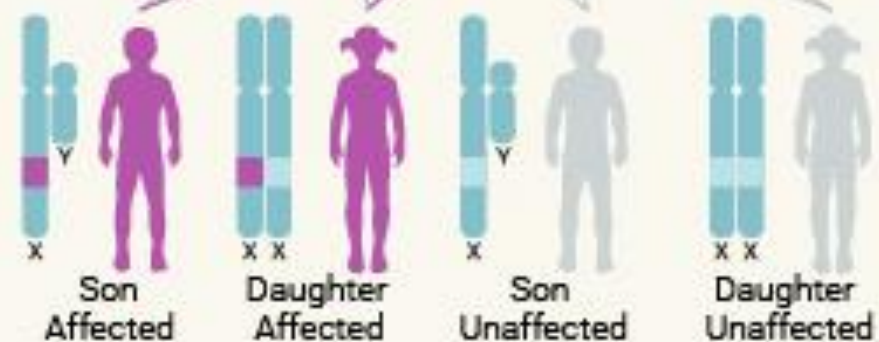
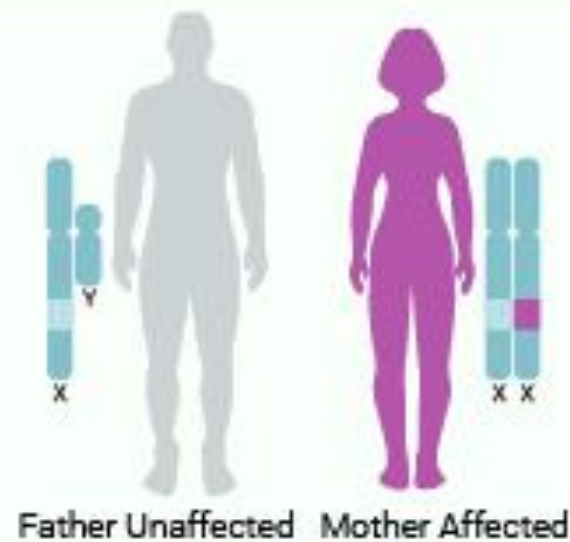
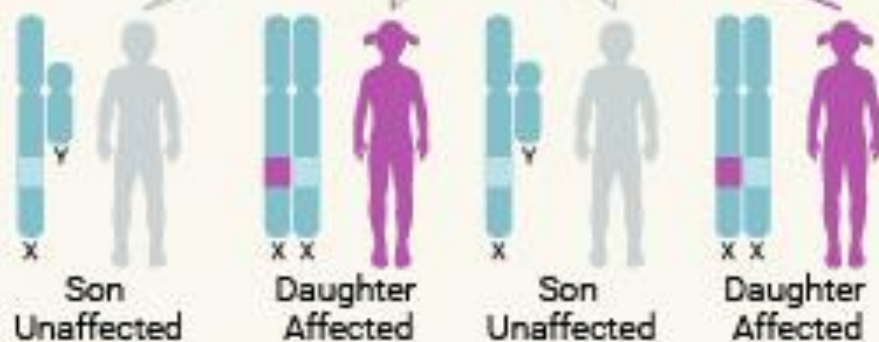


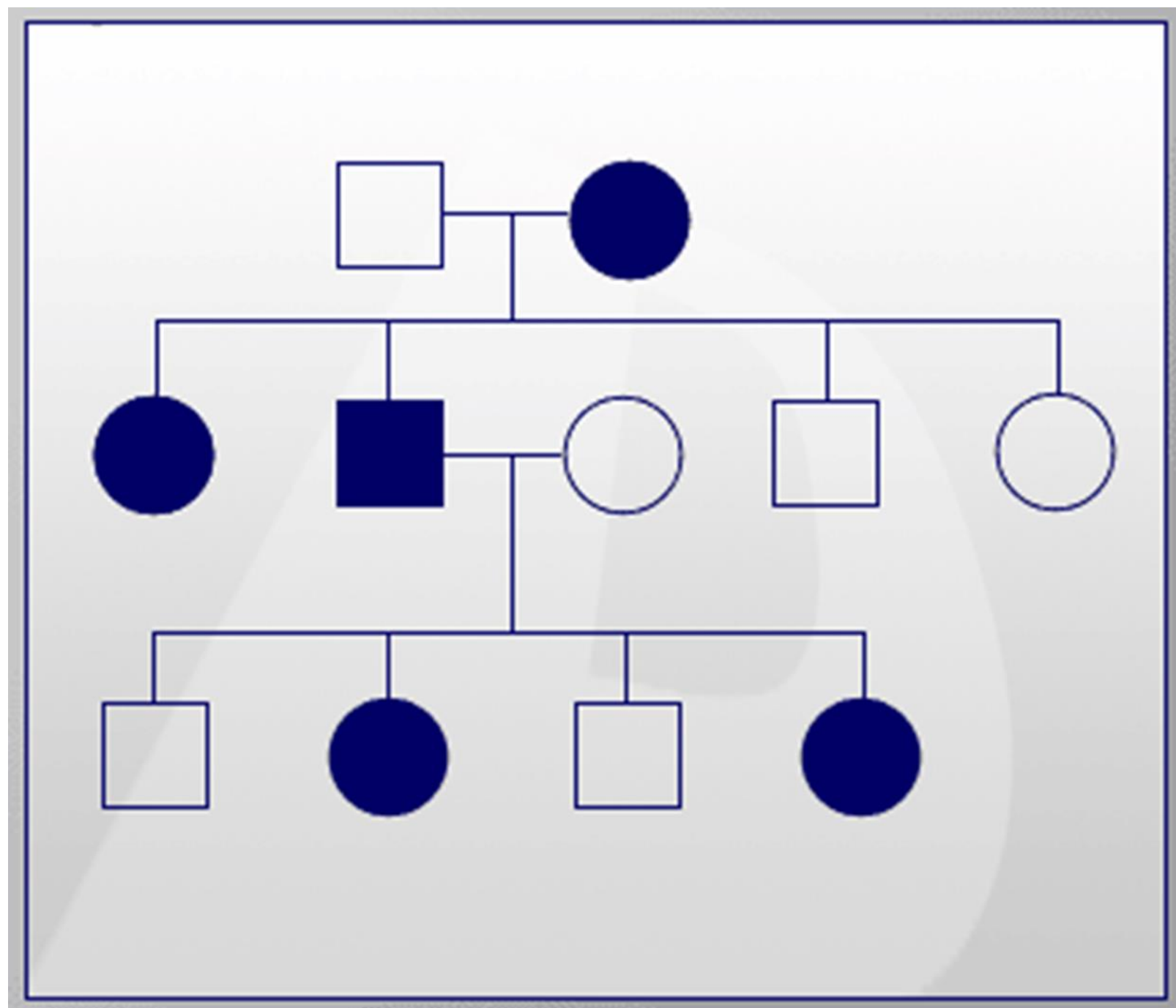
X-Linked Dominant

Parents



Children





Incontinentia Pigmenti XLDD

- Abnormal skin pigmentation, conical teeth and sometimes neurological signs seen only in female.

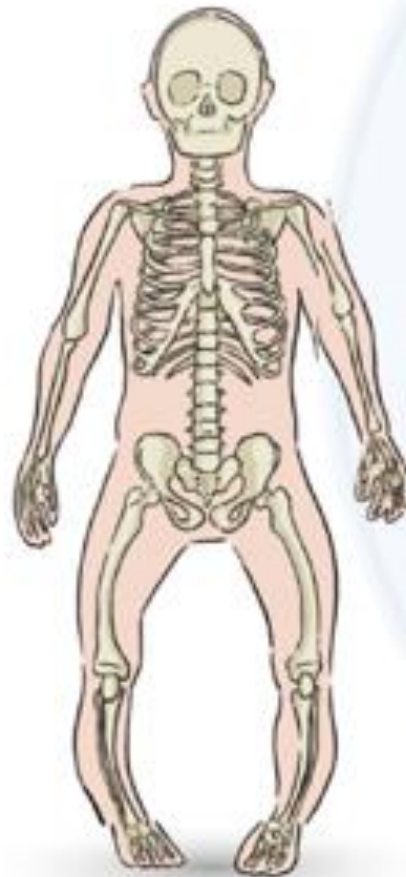


Hypophosphatemic Rickets

Vit D Resistance Rickets

- the kidneys cannot reabsorb phosphate effectively and too much of this mineral is excreted from the body in urine.
- As a result, not enough phosphate is available in the bloodstream to participate in normal bone development and maintenance result in abnormal ossification with bending and distortion of the bones.

PEDIATRIC



Delayed and disproportionate growth

Craniosynostosis

Rickets

Delayed motor development and gait abnormalities

Short stature

Deformity of weight-bearing limbs

Tooth abscesses

Excessive dental caries

Osteomalacia

Bone and joint pain

Joint stiffness

Muscle pain and weakness

Chiari malformation

Gait abnormalities

Diminished quality of life including psychosocial impact

ADULT



Fractures (including insufficiency fractures and Looser zones)

Osteoarthritis

Extrasosseus calcification including:

- Enthesiophytes
- Enthesopathy
- Spinal stenosis

Hearing loss

Disability that impacts ability to work

When and why females occasionally manifest features of X-linked recessive trait?

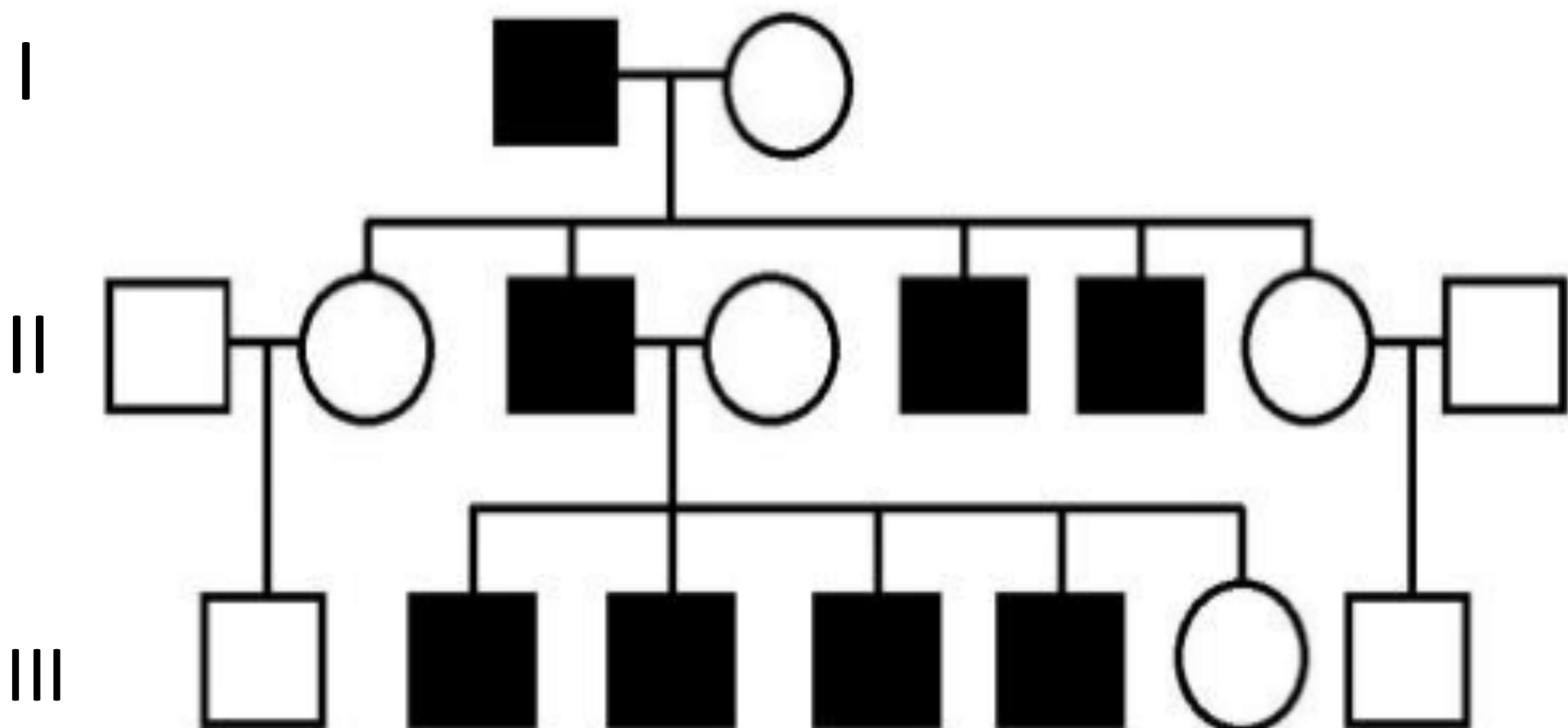
- 1-Homozygosity of X-linked recessive disorders
- 2-Numerical X-chromosome abnormalities(turner)
- 3-X-autosome translocation
- 4-Abnormal (skewed x) x-inactivation (not randomly inactivated)

Y chromosome contains relatively few dozen Y-linked (holandric) genes

- The gene that initiate differentiation of the embryo into male
- Several genes encodes testis-specific spermatogenesis factors

Y-Linked Inheritance

- Only males are affected
- Affected male transmits the trait to all of his sons and none of his daughters
- **Example:** hairy pinna(ears)

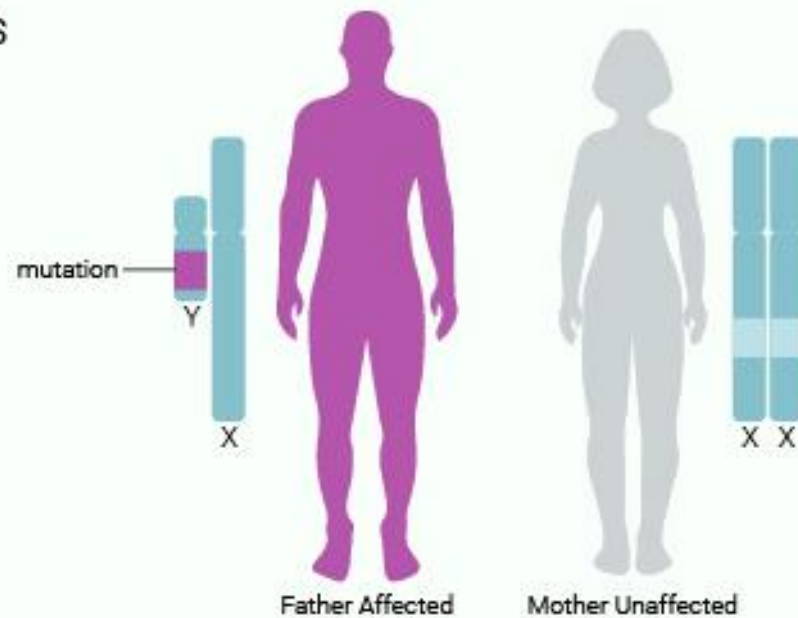




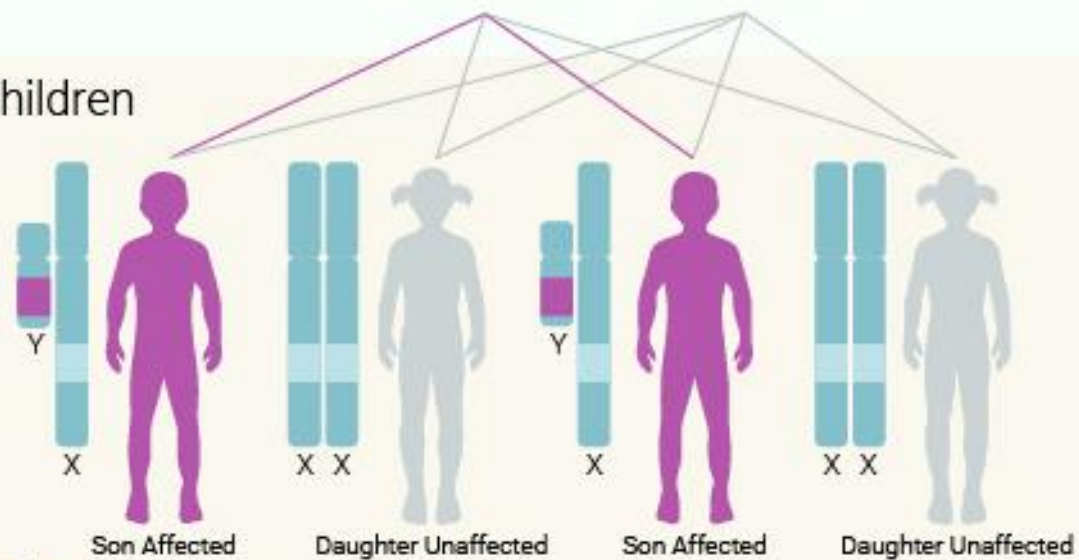
Hairy Pinna

Y-Linked

Parents



Children



Sex Influenced Traits

- Some autosomal traits are expressed more frequently in one sex than another
- E.g. gout (males are affected more than females)

Sex limited traits

- Occurs in only one of the sexes due to anatomical differences
- E.g inherited uterine or testicular defects.

Any
Question?

